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Master Thesis:

Federated Synthetic Data Analysis as a General Approach to Preserve Privacy and Data Structure: A Use Case Using Mixed Models

from

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Abstract

Federated data analysis enables collaborative research in situations where individual-level data cannot be shared across health institutions, for ethical, legal, and regulatory reasons. However, it often requires substantial computational infrastructure and coordination across all the sites. In this context, incorporating synthetic data generation into a federated analysis framework can provide a practical and cost-effective alternative by enabling centralised statistical modelling while still being able to preserve privacy. In this thesis, we evaluate the performance of federated synthetic data analysis for linear mixed effects models in a multi-site setting, assessing whether synthetic datasets generated within a federated analysis context can reproduce the inferential statistical results of gold standard pooled analyses while still maintaining low disclosure risk.

A simulated multi-site longitudinal dataset, motivated by real-world evidence settings, was used for evaluation. The data comprised of two repeated measurements per subject across eight sites and incorporated substantial residual variability, resulting in a low signal-to-noise ratio. Synthetic datasets were generated using variable-specific synthesis methods, including classification and regression trees for continuous variables and logistic regression for categorical variables. Statistical utility was assessed using propensity-score based measures, and federated synthetic data analysis inferential results were compared with those from gold standard pooled analyses across multiple repetitions. Disclosure risk was evaluated for both identity and attribute disclosure using established quasi-identifiers and sensitive target variables.

The results indicate that federated synthetic data analysis achieves good statistical utility, with synthetic data closely resembling the original data both in marginal and joint distributions. Fixed effects estimates were generally recovered with negligible bias, although bias was observed for categorical covariate, reflecting sensitivity to the choice of synthesizer. Differences were observed in subject-level and site-level variance estimates due to low signal-to-noise ratio induced by high residual variance and the small number of sites respectively. Despite these differences, overall inferential alignment between federated synthetic data analysis and gold standard pooled analysis was promising. Disclosure risks for both identity and attribute disclosure were consistently low across the sites.

In general, under the conditions investigated, these results demonstrate that federated synthetic data analysis provides a viable, privacy-preserving alternative to pooled multi-site analysis in complex data settings. By balancing analytical validity and confidential protection, federated synthetic data analysis offers a practical and cost effective alternative solution for collaborative research settings where data cannot be centrally shared, while highlighting important considerations for future research in federated synthetic data analysis.

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List of Abbreviations

Abbreviation	Definition
ADEMP	Aims, Data-generating mechanisms, Estimands, Methods, Performance measures
ARF	Adversarial Random Forests
BNs	Bayesian Networks
BHM	Bayesian Hierarchical Model
CART	Classification and Regression Trees
DataSHIELD	Data Aggregation Through Anonymous Summary-statistics from Harmonised Individual-Level Databases
Dorig	Disclosive in Original
Dsyn	Disclosive in Synthetic
DiSCO	Disclosive in Synthetic Correct Original
DiSDiO	Disclosive in Synthetic and Original
EHRs	Electronic Health Records
FDA	Federated Data Analysis
FSDA	Federated Synthetic Data Analysis
GDPR	General Data Protection Regulation
GLMMs	Generalized Linear Mixed Models
GLS	Generalized Least Squares
GANs	Generative Adversarial Networks
HIPAA	Health Insurance Portability and Accountability
IT	Information Technology
iSO	In Synthetic Original
LMMs	Linear Mixed-effects Models
MI	Multiple Imputation
MLE	Maximum Likelihood Estimation
MSE	Mean Squared Error
pMSE	Propensity Score Mean-Squared Error
RF	Random Forest
REML	Restricted Maximum Likelihood
RepU	Replicated Uniques
SD	Standard Deviation
SDG	Synthetic Data Generation
S_pMSE	Standardized Propensity Score Mean-Squared Error
SNR	Signal to Noise Ratio
UiO	Unique in Original

Abbreviation	Definition
UiS	Unique in Synthetic
UiOiS	Unique in Original and Synthetic
VAEs	Variational Autoencoders

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1 Introduction

1.1 Background of the Study

The sharing of health data is necessary for advancing health research, improving population health, informing public health policy, and driving data-driven innovation. Access to large, representative datasets enables researchers to uncover patterns and relationships that can inform public health interventions and support population-level health management. However, privacy concerns and strict data protection regulations, such as the General Data Protection Regulation (GDPR) in Europe, and the Health Insurance Portability and Accountability Act (HIPAA) in the United States, pose significant challenges to multi-jurisdictional data sharing and collaborative research efforts (Azizi et al. 2023).

To address these challenges, **Synthetic Data Generation (SDG)**, and **Federated Data Analysis (FDA)** have emerged as leading privacy-enhancing technologies. SDG involves creating artificial datasets that preserve the statistical properties of the original data without containing personal information, enabling data sharing while maintaining privacy (Azizi et al. 2023). In parallel, FDA enables decentralized data analysis across multiple sites without physically transferring the data, thereby maintaining compliance with privacy regulations (Azizi et al. 2023).

A hybrid approach, **Federated Synthetic Data Analysis (FSDA)**, combines SDG with a federated framework, producing synthetic datasets that can be securely pooled and analyzed while retaining key statistical characteristics of the source data. Recent advances in these technologies, such as **Adversarial Random Forests (ARF)** for generating high-utility synthetic data (Watson et al. 2023) and federated algorithms for **generalized linear mixed models (GLMMs)** (Yan et al. 2023), have expanded their potential for complex health research studies.

1.2 Problem Statement

Mixed-effects models provide a robust framework for analyzing complex hierarchical or clustered data (common in longitudinal and multi-center studies) by incorporating both fixed effects (population-level effects) and random effects (site-level or individual-level variability) (Yan et al. 2023). However, applying these models in an FSDA setting introduces unique challenges, such as:

- **Preservation of data structure**: Synthetic datasets must retain correlation structures and variance components critical for valid inference.
- **Statistical validity across sites**: Decentralized analyses must yield results comparable to pooled data approaches.
- **Privacy assurance**: Synthetic datasets must meet established confidentiality standards, such as low **membership disclosure risk**, i.e., the probability that an individual in the original dataset can be re-identified in the synthetic dataset (El Emam et al. 2022; Raab et al. 2024).

Despite the growing body of literature on SDG and FDA individually, systematic evaluations of **mixed-effects models under FSDA**, particularly when model complexity and inter-site heterogeneity vary, remains relatively under-explored.

1.3 Aims and Objectives

As part of this thesis, a simulation study was conducted to investigate the feasibility and validity of applying FSDA to mixed-effects models in a multi-site setting. In addition, privacy implications were evaluated using the synthetic datasets generated, although they were not measured across all simulation repetitions. Specifically, the objectives are to:

1. **Evaluate comparability**: Determine whether mixed-effects models fitted on FSDA datasets produce parameter estimates, standard errors, and variance components consistent with those from pooled original data analyses.
2. **Test robustness**: Examine the impact of increasing model complexity (e.g., additional random effects) and inter-site heterogeneity on FSDA performance.
3. **Assess privacy risk**: Quantify the membership disclosure risk of synthetic datasets to ensure compliance with confidentiality standards.

By addressing these objectives, this study provides methodological insights into the practical use of FSDA for epidemiological research and other fields involving sensitive, multi-source data, offering an alternative solution for privacy-preserving, multi-site data analysis.

2 Methodology

This chapter presents the methodological framework adopted in this thesis, focusing on the simulation study designed to evaluate FSDA in multi-site settings. The study is structured around the **ADEMP** framework: Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures, which provides a systematic approach for planning and reporting simulation studies, as described by Morris et al. (2019).

Our framework integrates FDA principles, SDG techniques, and mixed-effects modeling. By embedding these methods within a simulation study, we systematically evaluate the statistical validity of FSDA, while the privacy-preserving potential is assessed separately using the synthetic datasets.

The subsequent sections provide a description of the FSDA framework, the design of the simulation study, and the specific methods used for data synthesis, model fitting, and performance evaluation, including separate assessment of privacy-preserving characteristics.

2.1 Federated Data Analysis

FDA is a privacy-preserving approach to jointly analyze data from multiple sources without transferring individual-level data. In health research, it is particularly relevant where data sharing is restricted due to ethical, legal, or regulatory barriers, such as those imposed by the GDPR in the European Union or the HIPAA Act in the United States (Azizi et al. 2023).

Multi-center studies, especially those based on electronic health records (EHRs), registries, or international cohort data, are increasingly necessary to ensure increased sample size (statistical power), diversity of study population, and generalizability of the results. However, integrating individual-level data across institutions remains a challenge due to privacy concerns, large-scale computation resource requirements, heterogeneity across sites, and correlated observations (Yan et al. 2023).

2.1.1 Federated vs. Centralized Networks

Federated analysis enables scientific collaboration without violating local data governance rules. Instead of centralizing data, institutions retain their individual-level data locally and only share anonymous summary-level statistics, such as regression coefficients (Yan et al. 2023). This federated setup facilitates international collaboration while preserving data sovereignty and individual privacy. In contrast, in centralized networks, all individual-level data are pooled into a single repository for analysis. Although this allows for flexible model fitting, it may violate privacy regulations.

2.1.2 Theoretical Framework

Let K denote the number of participating data-holding institutions (or sites), and $\mathcal{D}_k$ denote the dataset held at site k, with n_k observations. Each observation in $\mathcal{D}_k$ can be represented as a tuple (row) containing all

measured variables:

$$\mathbf{x}_{ki} = (x_{ki1}, x_{ki2}, \dots, x_{kip}), \quad i = 1, \dots, n_k,$$

where p is the number of variables, and i indexes the observations at site k. Then, the full dataset and total number of observations across all sites is defined as

$$\mathcal{D} = \bigcup_{k=1}^{K} \mathcal{D}_k, \quad N = \sum_{k=1}^{K} n_k,$$

where:

- $\mathcal{D}$ is the full dataset, composed of the union of local datasets $\mathcal{D}_k = \{\mathbf{x}_{k1}, \mathbf{x}_{k2}, \dots, \mathbf{x}_{Kn_k}\}$,
- N is the total number of observations, as the sum of observations n_k from each site.

In centralized analysis, $\mathcal{D}$ is assembled physically before analysis - a process often infeasible under privacy constraints.

In FDA, the goal is to estimate a global model parameter θ by maximizing the log-likelihood function, denoted $\ell(\theta)$, which quantifies how well the parameter θ explains the observed data, without pooling or centralizing the individual-level data ($\mathcal{D}_k$) due to data privacy and regulatory restrictions that prohibit centralization. Each site k contributes to the total model log-likelihood via its **local likelihood function**, denoted $\ell_k(\theta)$ (Yan et al. 2023). Assuming independence of observations between different sites, the **global log-likelihood** can be written as the sum of local-likelihoods (Yan et al. 2023):

$$\ell(\theta) = \sum_{k=1}^{K} \ell_k(\theta)$$

Federated analysis solves the centralization issue by operating on **local likelihood functions** (site-specific likelihood functions) $\ell_k(\theta)$ instead of the pooled global likelihood (Yan et al. 2023). Each site computes local components of the likelihood, which are summarized as summary statistics, denoted as s_k, which are functions of:

- The local data $\mathcal{D}_k$
- The current global parameter θ
- Any site-specific parameter δ_k

$$s_k = s_k(\mathcal{D}_k, \theta, \delta_k)$$

Once all sites transmit their s_k to the central analysis node, the central server can aggregate and update the parameter estimate $\hat{\theta}$ by maximizing the sum of local log-liklihoods $\ell_k(\theta)$ from all K sites (Yan et al. 2023). The estimator is then obtained by solving:

$$\hat{\theta} = \arg\max_{\theta} \sum_{k=1}^{K} \ell_k(\theta)$$

This approach enables the central site to perform iterative optimization as if it had direct access to all data, despite only receiving aggregate information.

2.1.3 Real-World Applications

FDA has proven valuable across a wide range of disciplines where data are distributed across institutions and cannot be directly shared due to privacy, legal, or ethical reasons. Its adoption has been particularly impactful in healthcare, epidemiology, public health, and biomedical informatics, where multi-source data integration is essential for robust and generalizable findings.

The key application areas of FDA includes:

- **Multi-center clinical trials**, where patient-level data are collected at different hospitals or research centers, but data protection regulations often prevent direct pooling.
- **Population-level health registries**, used for surveillance and outcomes research across jurisdictions, where legal and governance restrictions limit central sharing.
- **Epidemiological studies**, particularly large-scale cohort and case-control studies where privacy and ethical safeguards prohibit direct exchange of participant-level data.
- **Genomic and molecular research**, where privacy risks and re-identification concerns impose strict data-use conditions.
- **Pharmacovigilance and drug safety**, involving confidential EHR and claims data across health systems, where institutional and regulatory barriers must be respected.
- **International health collaborations**, such as those involving cross-border cohorts or public health emergencies.

2.1.4 DataSHIELD as an FDA Implementation

One widely adopted FDA infrastructure is **DataSHIELD** (Data Aggregation Through Anonymous Summary-statistics from Harmonised Individual-levEL Databases), which offers infrastructure and software for performing privacy-preserving statistical analyses across distributed datasets. It is a distributed approach that

allows the analysis of sensitive individual-level data from one study, and the co-analysis of such data from several studies simultaneously without physically pooling them or disclosing any data (Wilson et al. 2017).

DataSHIELD operates through a structured workflow, which can be summarised in the following steps:

1. The researcher (analyst) submits an analysis request from a central client node to each participating site.
2. **Local servers** (behind institutional firewalls) execute the requested analysis code on their local dataset.
3. Only **summary-level outputs** (e.g., regression coefficients, aggregated counts, test statistics) are returned to the central node.
4. The **central node** aggregates these summaries to produce global results without ever accessing raw data.

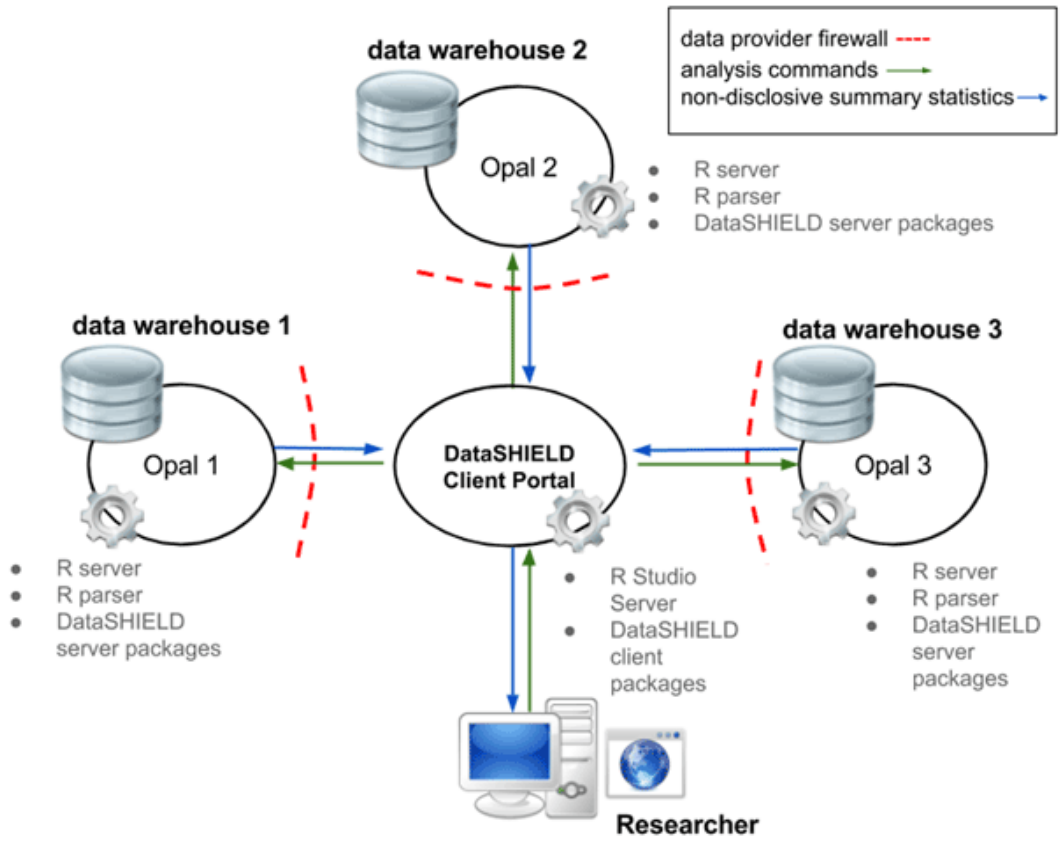


Figure 2.1: *Schematic representation of the DataSHIELD infrastructure. Researcher initiated analysis commands (green arrows) are sent to each participating data warehouse, where local R servers execute the analysis behind institutional firewalls (red dashed lines). Only non-disclosive summary statistics (blue arrows) are returned to the central client portal for aggregation and inference. Sourced from: Wilson et al. (2017).*

This workflow offers several advantages:

- **Privacy compliance:** Adheres to data protection regulations such as GDPR, HIPPA, and local governance rules.
- **Data sovereignty:** Eliminates the need for secure transfer of individual-level data.
- **Flexibility:** Supports a wide range of statistical models, including regression and generalized linear models.

Nevertheless, some limitations remain:

- **Infrastructure and setup costs:** Requires installation and configuration of compatible server-side software (e.g., Opal, R packages) at all sites, often involving collaboration with local information technology (IT) teams. This can be time-consuming and require significant resources (Wilson et al. 2017).
- **Maintenance burden:** Sites must coordinate on version control, security patches, and software updates to ensure compatibility across the network.
- **Communication overhead:** Iterative algorithms (e.g., model fitting with multiple optimization steps) may require repeated back-and-forth communication, increasing latency and processing time.

2.2 Synthetic Data Generation

2.2.1 Definition and Purpose

SDG refers to the process of replacing observed values in a dataset with simulated values drawn from appropriate probability distributions, in such a way that the statistical properties, relationships, and structure of the original data are preserved (Nowok et al. 2016; Reiter 2005a). This replacement may involve **all variables** (full synthesis) or only a **subset of variables** (partial synthesis). Partial synthesis is often applied when only certain variables are sensitive and need protection, while others can remain untouched. The most primary purpose of SDG is to enable data sharing and analysis while protecting respondent confidentiality, thereby ensuring compliance with privacy regulations such as GDPR and HIPPA.

Unlike FDA, which keeps data stored at local sites and performs computation remotely, SDG produces stand-alone datasets that can be freely shared and centrally analyzed after generation, reducing the need for specialized secure infrastructures.

2.2.2 Evolution of Synthetic Data Generation Methods

2.2.2.1 Roots in Multiple Imputation

The theoretical foundation for SDG originates in the framework of **Multiple Imputation (MI)**, first proposed by Rubin (1987) and extended to confidentiality protection by Little (1993) and Rubin (1993). This framework was further formalized by Raghunathan et al. (2003).

The central idea is to treat unobserved or un-sampled units as **missing data** and to generate multiple simulated values by imputation from appropriate predictive distributions. These multiple completed datasets are then analyzed separately, with results combined to reflect the additional uncertainty introduced by imputation. While MI imputes only missing values (leaving the observed ones intact), SDG extends this principle by replacing observed values of sensitive variables in order to prevent disclosure. Conceptually, SDG can be applied in two ways: **fully synthetic data**, in which all values of sensitive variables are replaced (Little 1993; Rubin 1993), and **partially synthetic data**, in which only certain variables are replaced (Raghunathan et al. 2003; Reiter 2005a). In this thesis, the focus is only on fully synthetic data to illustrate the principles and methodology of SDG while ensuring privacy protection in multi-site data analyses.

2.2.2.2 Early Parametric Synthesis

Early implementations of SDG primarily employed **parametric models** (Reiter 2003 ; Rubin 1993). These methods involved specifying distributional forms (e.g., normal for continuous data, logistic for binary outcomes), estimating model parameters from observed data, and generating synthetic values either by **simple synthesis**, which draws synthetic values directly from the fitted model or by **proper synthesis**, which accounts for parameter uncertainty by first sampling parameter values from their posterior distribution and then drawing synthetic data from the corresponding posterior predictive distribution. This approach ensures that both the uncertainty in the model parameter and the variability in the data are reflected in the synthetic datasets. In this thesis, we adopt the proper synthesis approach to ensure realistic data generation and valid downstream inference in the simulation study.

The advantage of parametric synthesis lies in its strong Bayesian theoretical foundation. However, these methods can perform poorly when model assumptions are violated or when data exhibits complex nonlinearities and interactions (Reiter 2003; Rubin 1993).

2.2.2.3 Non-Parametric Methods

To address the limitations of parametric approaches, later work incorporated **non-parametric models** (machine learning models) into the SDG process. For example, Reiter (2005b) proposed the use of Classification and Regression Tress (CART), while Caiola and Reiter (2010) extended this to Random Forests (RF). These approaches avoid strict distributional assumptions and are well-suited to datasets with mixed data types and complex dependencies (Reiter 2005b).

The **synthpop** package in R which was incorporated into this thesis, integrates both parametric and non-parametric synthesis methods, including CART and RF, and implements SDG using a **sequential conditional modeling** framework (Nowok et al. 2016).

2.2.2.4 Sequential Conditional Modeling

In this approach, the joint distribution of the dataset is represented as a sequence of conditional distributions. First, a variable is selected, and its conditional distribution is estimated given a set of predictor variables.

Next, the following variable is modeled conditional on both the available predictors and the already synthesized variables. This process is repeated sequentially until all variables in the dataset have been synthesized (Nowok et al. 2016).

2.2.3 Theoretical Framework

2.2.3.1 Rubin's Contribution (1993)

Rubin (1993) proposed a framework for fully synthetic data generation based on the multiple imputation principle. His idea was to treat the entire population as "missing" and impute values for all units from the posterior predictive distribution conditional on observed data, thereby generating synthetic datasets that preserve the statistical properties of the original data. Formally, Rubin conceptualized the model as:

$$Y_{syn} \sim f(Y \mid X, \theta),$$

where θ is a vector of model parameters, and Y_{syn} is the synthetic data generated from the posterior predictive distribution of Y given X and model parameters θ. This approach assumes a parametric model for $Y \mid X$ and allows the generation of synthetic datasets reflecting both model-based uncertainty and sampling variability.

In our simulation study using `synthpop`, we generate synthetic data directly from the observed sample rather than a full conceptual population. Therefore, we write:

$$Y_{syn} \sim f(Y_{obs} \mid X, \theta),$$

This resulting complete synthetic population reflects both imputation and population uncertainty. Since releasing entire synthetic populations may be infeasible, a random sub-sample of synthetic records can be released instead.

This ensures that no original records are disclosed, while maintaining valid statistical properties for downstream analysis.

2.2.3.2 Proper vs. Simple Synthesis

In synthetic data generation, the observed data Y_{obs} are assumed to be sample from a population with an underlying joint distribution, which can be expressed in terms of conditional distributions. The synthesizer, which is the algorithm or model used to generate synthetic data, produces synthetic values for each variable sequentially conditional on the observed covariates X and previously synthesized columns of Y_{obs}. This sequential approach preserves the multivariate structure of the original data.

Synthetic datasets can be generated using either **simple synthesis** or **proper synthesis**, depending on whether parameter uncertainty is taken into account:

Simple synthesis generates synthetic data by drawing directly from the fitted conditional distributions of the observed data without accounting for parameter uncertainty. The model parameters θ are estimated once from the observed data and treated as fixed when generating synthetic data. Formally, for the m-th dataset, the synthetic values are generated as

$$Y_{\text{syn}}^{(m)} \sim f(Y_{\text{syn}} \mid X, Y_{\text{obs}}),$$

where $Y_{\text{syn}}^{(m)}$ denotes the synthetic values for the m-th dataset, and $m = 1, \dots, M$ indexes the number of synthetic datasets produced. Because parameter uncertainty is ignored, this approach may underestimate variability in downstream inference (Nowok et al. 2016).

Proper synthesis accounts for parameter uncertainty by first sampling model parameters from their posterior distribution,

$$\theta^{(m)} \sim p(\theta \mid X, Y_{obs}),$$

and then generating synthetic values from the corresponding posterior predictive distribution,

$$Y_{\text{syn}}^{(m)} \sim f(Y_{\text{syn}} \mid X, \theta^{(m)}),$$

where $Y_{\text{syn}}^{(m)}$ denotes the synthetic values for the m-th dataset. This procedure ensures that both sources of uncertainty, the uncertainty in the estimated parameters (θ) and the inherent variability in the data, are propagated into the synthetic datasets. As a result, the released data provide more valid inference by properly reflecting the additional variability introduced through the synthesis process (Nowok et al. 2016).

In practice, proper synthesis is the preferred approach, as it accounts for both parameter uncertainty and the inherent variability in the data. It is the method implemented in many sythetic data tools, such as the `synthpop` R package (Nowok et al. 2016), and it is the approach used in this thesis.

Since releasing entire synthetic populations may be infeasible, Rubin (1993) suggested releasing random sub-samples of synthetic records instead. This guarantees that no original records are disclosed while maintaining valid statistical properties for inference.

2.2.3.3 Raghunathan et al. (2003) - The Two-Step Procedure

Raghunathan et al. (2003) formalized Rubin's conceptual idea into a practical **two-step procedure** for constructing synthetic datasets. The first step corresponds directly to what we have described in Section 2.2.3.2 as proper synthesis. Their key contribution was to add a second step, in which synthetic datasets are sampled from the completed synthetic population, introducing an additional layer of randomness and making the framework distinct from standard multiple imputation. The procedure is described as follows:

1. **Posterior Predictive Imputation**: Considering a population of size N. Let $z_j = 1$ if unit j is observed in the original sample, and $z_j = 0$ otherwise, so that $\mathbf{z} = (z_1, \dots, z_N)$ is the vector of sample inclusion indicators. This partition implies

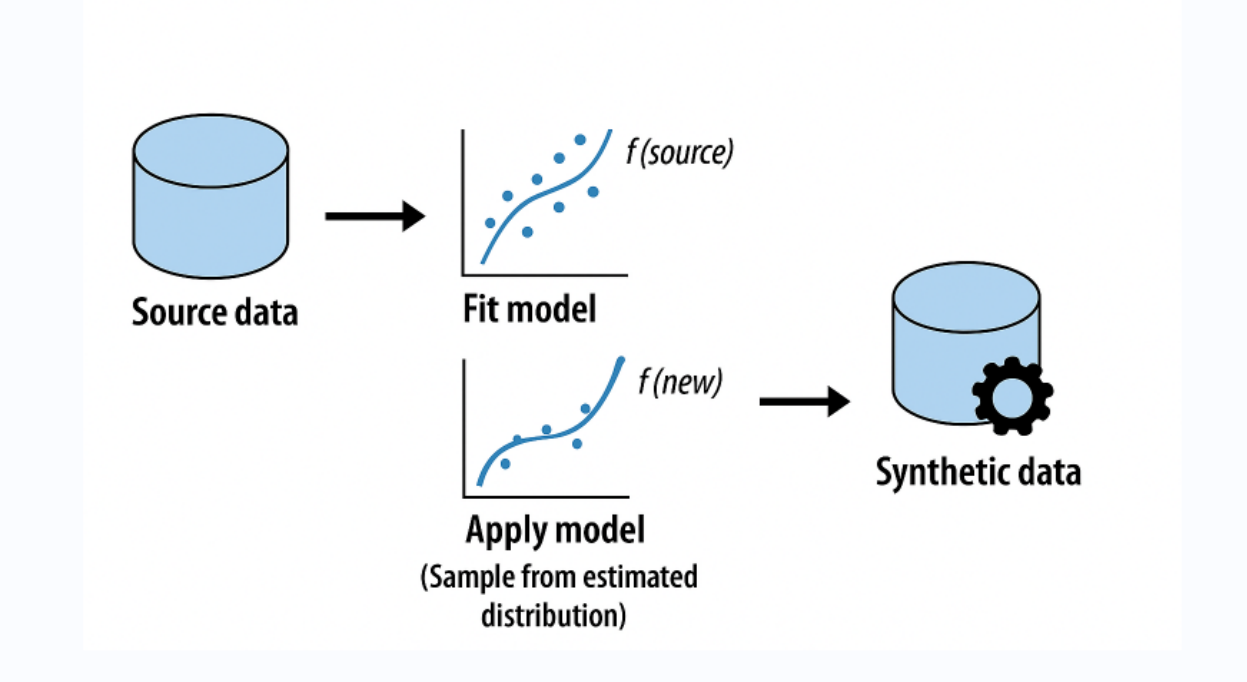


Figure 2.2: *Conceptual illustration of synthetic data generation. A model is first fitted to the original ("source") data, and synthetic values are then generated by sampling from the estimated distribution implied by the model, producing synthetic datasets that preserve the statistical structure of the source data. Source: adapted from El Emam et al. (2020).*

- Y_{obs}: observed values for sampled units $z_j = 1$
- Y_{nobs}: unobserved values for non-sampled units $z_j = 0$
- X: design variables available for all N units (e.g., strata, cluster, or size measures)

Given observed data $(X, Y_{\text{obs}}, \mathbf{z})$, the imputer (a statistical model or algorithm that generates synthetic or missing values - in synthpop, this is implemented through methods such as CART, logistic regression, or RF) produces the unobserved or un-sampled values Y_{nobs} from the posterior predictive distribution (Nowok et al. 2016; Reiter 2005a)

$$Y_{\text{nobs}} \sim f(Y_{\text{nobs}} \mid X, Y_{\text{obs}}, z).$$

This results in a completed synthetic population $(X, Y_{\text{com},i})$, where the subscript i indexes the i-th completed population. Because the imputation is repeated m times independently, each $Y_{\text{com},i}$ is slightly different, reflecting the uncertainty from the posterior draws.

In this theoretical setup, $\mathbf{z}$ plays a role in distinguishing sampled from non-sampled in the full population. However, in practical implementation such as the `synthpop` package in our simulation study, the synthesis is performed only on the sampled dataset itself, without explicit use of $\mathbf{z}$. We replace

$$Y_{\text{nobs}} \sim f(Y_{\text{nobs}} \mid X, Y_{\text{obs}}, z)$$

with

$$Y_{\text{syn}} \sim f(Y_{\text{syn}} \mid X, Y_{\text{obs}}, z)$$

to indictae that in our implementation, the imputer (synthesizer) generates synthetic datasets rather than imputing unobserved population values. The z is included here to remain consistent with original formulation by Raghunathan et al. (2003), but for our empirical work it is not directly relevant and therefore excluded from implementation. In this thesis, the *imputer* (synthesizer) corresponds to the sequential regression modeling engine in `synthpop`, which generates one variable at a time conditional on the previously synthesized variables. This ensures that the multivariate structure of the original data is preserved while introducing variability through proper synthesis (Nowok et al. 2016; Reiter 2005a).

2. **Synthetic Sampling**: From each completed population $(X, Y_{\text{com},i})$, draw a sample of size n_{syn} to form one synthetic dataset $(X, Y_{\text{syn},i})$. Repeating this process for $i = 1, \dots, m$ yields m multiply imputed synthetic datasets.

This framework introduces two layers of randomness: first, through both the imputation of unobserved (synthetic) values drawn from estimated conditional distributions, and second, through resampling from the completed synthetic populations. This makes the inferential framework distinct from that of standard multiple imputation (Raghunathan et al. 2003).

2.2.3.4 Inference and Combining Rules

Let Q denote an estimand of interest (e.g., a population mean or regression coefficient). From each synthetic dataset i, the analyst computes an estimate q_i and its estimated variance v_i, for $i = 1, \dots, m$.

Fully Synthetic Data

In the standard framework for fully synthetic data generation, the synthesizer produces $m > 1$ synthetic datasets. The analyst then fits the desired statistical model separately to each of these datasets, and the resulting estimates are combined in order to account for two sources of variability. The first source is the within-synthesis variance, denoted $\bar{v}$, which captures the average model-based variance from each synthetic dataset. The second source is the between-synthesis variance, denoted b, which captures the variability of parameter estimates across the m synthetic datasets. More formally, if q_i denotes the point estimate of the parameter of interest Q obtained from the i-th synthetic dataset, for $i = 1, \dots, m$, and v_i represents the estimated variance of q_i from the i-th synthetic dataset, then the final inference is based on combining information from all m number of synthetic datasets.

The combined point estimate is the simple average:

$$\bar{q}_m = \frac{1}{m} \sum_{i=1}^{m} q_i$$

The average within-synthesis variance is:

$$\bar{v}_m = \frac{1}{m}\sum_{i=1}^{m} v_i$$

The between-synthesis variance is:

$$b_m = \frac{1}{m-1}\sum_{i=1}^{m}(q_i - \bar{q}_m)^2$$

The total variance, for fully synthetic data; accounting for both within- and between-synthesis variability, is:

$$T_s = \left(1 + \frac{1}{m}\right) b_m - \bar{v}_m$$

A large-sample 95% confidence interval for Q is then given by:

$$\bar{q}_m \pm 1.96\sqrt{T_s}$$

These combining rules, originally proposed by (Reiter 2005a), ensure that additional uncertainty introduced by the synthesis process is properly reflected in statistical inference.

Partially Synthetic Data

When only some variables in the dataset are replaced by synthetic values, the variance is estimated differently (Reiter 2003; Rubin 1993). In this case, the total variance is defined as

$$T_p = \bar{v}_m + \frac{b_m}{m},$$

and the corresponding 95% confidence interval for Q is

$$\bar{q}_m \ \pm \ 1.96\sqrt{T_p}.$$

Thus, while both approaches use multiple synthetic datasets and combine estimates across them, the combining rules differ because fully synthetic datasets add an additional layer of uncertainty from resampling entire populations.

2.3 Federated Synthetic Data Analysis (FSDA)

FSDA is a novel framework that integrates ideas from FDA and SDG to enable multi-site statistical modeling without requiring the sharing of original, sensitive individual-level data. Traditional FDA ensures that original data never leave the local site, while SDG enables data sharing by replacing original records with synthetic versions.

In FSDA, each participating site retains its original dataset and performs **local synthetic data generation**. This step replaces identifiable records with statistically similar but non-identifiable synthetic records, ensuring that no confidential information is shared outside the site.

Once local synthesis is complete, the synthetic datasets from all sites are **pooled centrally** into a single synthetic dataset. This pooled dataset can then be analyzed using the desired statistical model. In this simulation study, we employed a **linear mixed-effects model** to account for both site-level heterogeneity and repeated measurements within subjects.

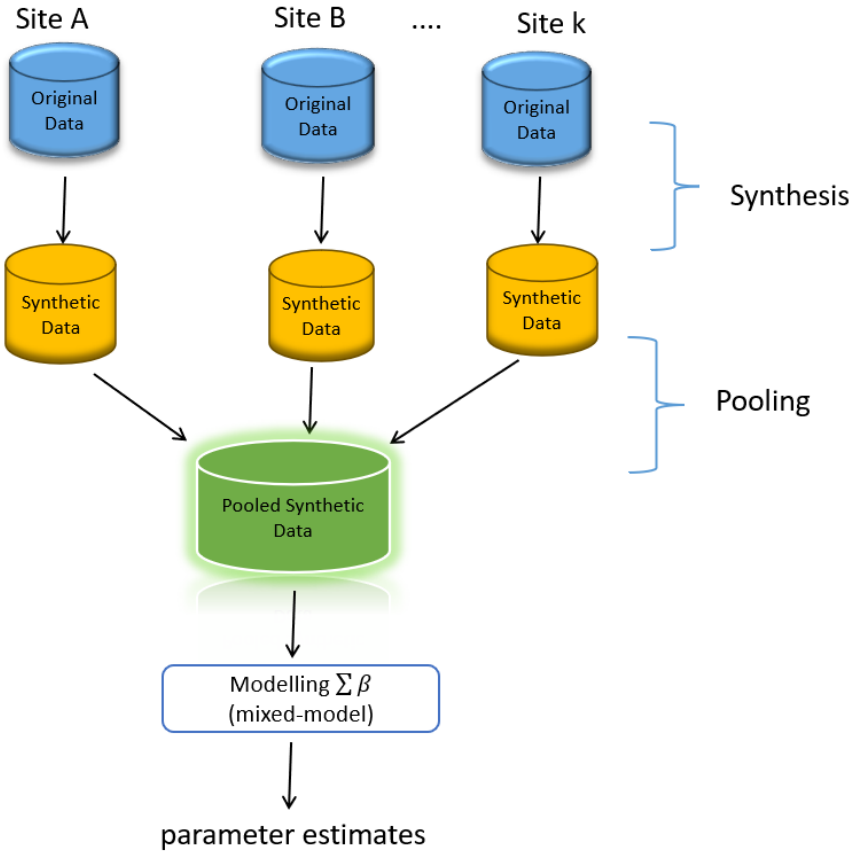


Figure 2.3: *Illustration of the FSDA process. At each participating site, the original dataset (blue) is converted into a synthetic dataset (yellow) through a chosen synthesis method. The synthetic datasets from all sites are then pooled centrally (green) and analyzed using a mixed-effects model. This approach allows for collaborative statistical modeling while preserving data privacy by ensuring that no original individual-level data leaves the local site.*

This process enables collaborative statistical modeling while enhancing privacy, as the original data remain at each site. Synthetic data are generated locally and subsequently pooled for joint analysis. Hierarchical structures, such as site-level clustering and repeated measurements within subjects, may be preserved through appropriate synthetic strategies, including the use of flexible modeling approaches (e.g., CART, random forest or parametric models), specification of predictor relationships, and techniques such as stratification or nested synthesis (Nowok et al. 2016). Appropriate analysis models, such as mixed-effects models, can then

be applied to the pooled synthetic data to estimate fixed and random effects.

2.3.1 Operational Steps in FSDA

The FSDA process can be described in three stages:

1. **Local Synthesis**

- Each site $k = 1, \dots, K$ begins with its original dataset (X_k, Y_k).
- Using a chosen synthetic method (e.g., parametric regression models, CART, or Random Forests), the site generates one or more synthetic datasets (X_s^*, Y_s^*).
- Sensitive identifiers are replaced with simulated values generated from statistical model fitted to the original data (e.g., parametric regression, CART, or Random Forest), with the aim of capturing the statistical distribution and relationships observed in the local data.

2. **Central Pooling**

- Synthetic datasets from all sites are combined into a single multi-site synthetic dataset. Each site contributes its locally synthesized data, which are then concatenated (pooled) to form:

$$(X^*, Y^*) = \bigcup_{k=1}^{K} (X_k^*, Y_k^*)$$

- Because the synthesis is performed site-wise, the hierarchical structure (e.g., subjects nested within sites) is retained in the pooled synthetic data.

3. **Analysis of Pooled Synthetic Data**

- The pooled dataset is analyzed using the same statistical models as would be applied to the original combined data.
- In this simulation study, we applied the **linear mixed-effects model** described in Section 2.5.3, which includes both site-level and subject-level random effects.

These key steps in the FSDA framework are implemented in this thesis are and visually summarized in Figure 2.3.

2.4 CART-Based Synthesis within FSDA

2.4.1 Overview and Notation

Following the framework described by Drechsler and Reiter (2011) and Reiter (2005c), CART was used as the primary synthesizer for continuous variables within the FSDA framework. The synthesis process is first described for a single variable and then extended sequentially to all variables subject to synthesis.

Let Y_i denote the variables to be synthesized and Y_{-i} the set of all remaining variables. For each record $j = 1, ..., n$, let $Z_j = 1$ indicate that the value of Y_{ij} is to be replaced by a synthetic value, and $Z_j = 0$ otherwise. The indicator vector $Z = (Z_1, ..., Z_n)$ defines the subset of records subject to synthesis.

2.4.2 CART as a Conditional Distribution Model

CART approximate the conditional distribution of a univariate outcome given a set of predictors by recursively partitioning the predictor space into subsets with relatively homogeneous outcome values (Breiman et al. 2017). The resulting partitions can be represented as a tree structure, where each terminal node (leaf) corresponds to a region of predictor space and defines a local conditional distribution of the outcome Y_i.

In the context of synthetic data generation, CART is used to approximate

$$p(Y_i|Y_{-i}),$$

without imposing any parametric assumptions on the function. This flexibility makes CART vary much suitable for settings with complex interactions and non-linear relationships (Drechsler and Reiter 2011).

2.4.3 Tree Construction for Synthesis

To construct the synthesis tree $Y^{(i)}$, only records with $Z_j = 1$ are used, ensuring that the tree is tailored to the subset of data to be replaced (Drechsler and Reiter 2011).The tree is grown by recursively splitting the predictor space Y_i such that:

- For continuous outcomes, splits are chosen to minimise the within-node deviance of Y_i;
- For cartegorical outcomes, splits minimise the Gini impurity.

Tree growth is subject to two constraints. First, each terminal leaf must contain at least k observations, with $k = 5$ used in this thesis, consistent with common practice and prior empirical evaluations (Drechsler and Reiter 2011). Second, splitting stops when further partitioning fails to reduce within-leaf deviance below a pre-specified threshold or when further partitioning or splitting would produce child leaves smaller than the minimum size $k = 5$.

2.4.4 Generation of Synthetic Values via Bayesian Bootstrap

For each record with $Z_j = 1$, the fitted tree $Y^{(i)}$ is traversed according to its predictor values until the corresponding terminal leaf is reached. Let L_w denote the w-th terminal leaf, and let

$$Y^{(i)}_{L_w} = \{Y_{ij} : j \in L_w\}$$

denote the observed values of Y_i within that leaf.

Synthetic values are generated by sampling from $Y^{(i)}_{L_w}$ using the Bayesian bootstrap (Rubin 1981). This procedure preserves the empirical distribution within each leaf while introducing stochastic variability, thereby avoiding direct disclosure of original values. Repeating this process for all records and across m iterations yields m partially synthetic datasets.

2.4.5 Confidentiality Enhancements

As discussed by Reiter (2005c) and Drechsler and Reiter (2011), additional confidentiality protection can be achieved through pruning of the CART tree. In particular, trees may be pruned to ensure sufficient diversity of values within leaves prior to resampling. For sensitive continuous variables, the empirical distribution within a leaf may be smoothed using a Gauassian kernel density estimator, and synthetic values drawn from the estimated density via inverse cumulative distribution function sampling. These extensions reduce the probability of releasing values too close to the original observations.

2.4.6 Extension to Full Synthesis and Implementation in `synthpop`

The synthesis framework described above follows the general principles outlined by Reiter (2005c) and Drechsler and Reiter (2011), which are primarily presented in the context of partially synthetic data. In this thesis, and as mentioned previously, a fully synthetic data approach as described by Rubin (1987) and Raghunathan et al. (2003) in Section 2.2.3 was adopted, in which all analysis variables were sequentially synthesized.

Full synthesis was implemented using the `synthpop` package, which extends the CART based synthesis framework to a sequential conditional modelling strategy. Variables were synthesized one at a time, with each variable conditioned on a combination of previously synthesized variables and remaining original variables, consistent with the joint distribution approximation framework described by Reiter (2005c). This sequential conditioning allows dependencies among variables to be preserved while ensuring that the final released datasets contain no original values for the synthesized variables.

CART was used as the primary synthesizer for continuous variables, while categorical variables were synthesized using logistic regression models. This approach was specifically chosen to bring about a balance between flexibility and model simplicity, and is consistent with common practice in synthetic data generation workflows in the `synthpop` package.

CART models were implemented using the default settings of the `synthpop` package. No manual tree pruning or custom confidentiality constraints were applied beyond these defaults. Smoothing of continuous variables within terminal node was handled internally by the package, which applies spline smoothing to the empirical distributions where appropriate and helps in further reducing disclosure risk.

2.5 Linear Mixed-Effects Models

Linear Mixed-effects Models (LMMs), also known as hierarchical or multilevel models, extend standard linear regression by including both **fixed effects** (population-level parameters) and **random effects** (subject- or cluster-specific deviations). They are particularly suited for analyzing **clustered**, **longitudinal data**, **repeated measures data**, and **multilevel data** where observations within subjects (or groups) are correlated and standard regression assumptions of independence are violated (Laird and Ware 1982; Pinheiro and Bates 2004a; Verbeke 2001).

2.5.1 Model Motivation

In this simulation study, repeated vitamin D level measurements are collected for subjects across multiple sites. Observations within a subject are correlated due to repeated measures, and observations within the same site may also be correlated due to shared context or conditions. Ignoring this correlation structure can lead to underestimated standard errors and invalid inference.

LMMs address this by introducing **subject-specific random effects** (to account for repeated measures or longitudinal data) and **site-specific random effects** (to account for clustering by site), while still estimating population-level fixed effects such as age, sex and time outdoors.

2.5.2 General Model Specification

Following the formulation of Verbeke (2001), a general LMMs for longitudinal data can be written as:

$$Y_i = X_i\beta + Z_i\mathbf{b}_i + \varepsilon_i,$$

with

$$\mathbf{b}_i \sim \mathcal{N}(\mathbf{0}, D), \quad \varepsilon_i \sim \mathcal{N}(\mathbf{0}, \Sigma_i), \quad \mathbf{b}_1, \ldots, \mathbf{b}_N, \varepsilon_1, \ldots, \varepsilon_N \text{ independent},$$

where:

- Y_i is the vector of repeated measurements for subject i,
- X_i is the design matrix for fixed effects,
- β is the vector of fixed effect coefficients (population-level parameters),
- Z_i is the design matrix for random effects,

- $\mathbf{b}_i$ are subject-specific random effects, assumed multivariate normal with covariance D,
- ε_i are residual errors with covariance Σ_i.

This formulation gives a general view of mixed-effects structure and allows for multiple covariates in X_i.

2.5.3 LMM for this Simulation Study

To analyze the pooled synthetic dataset, we specify a **three-level Linear Mixed-effects Models (LMMs)** that accounts for repeated measurements (i) within subjects (j) and nested within sites (k), we define the indexing scheme as:

- $i = 1, \dots, n_j$: repeated measurements within subject j (Level 1)
- $j = 1, \dots, N_k$: subjects nested within site K (Level 2)
- $k = 1, \dots, K$: sites (Level 3)

Thus, the outcome variable is Y_{ijk}, representing the i-th measurement for subject j in site k.

The LMM for the pooled data (original or synthetic) is:

$$Y_{ijk} = \beta_0 + \beta^\top \mathbf{X}_{ijk} + u_k + u_{j[k]} + \varepsilon,$$

where:

- $\mathbf{X}_{ijk}$ is a vector of covariates,
- β is the vector of fixed-effect coefficients $(\beta_1, \beta_2, \beta_3, \dots)$,
- $u_k \sim \mathcal{N}(0, \sigma_u^2)$ is the site-specific random intercept,
- $u_{j[k]} \sim \mathcal{N}(0, \sigma_{u_j}^2$ is the subject-specific random intercept nested within site k,
- $\varepsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ is the residual error (within-subject variation).

This model specification allows us to simultaneously capture variability at both site level and the subject level, while estimating population-level fixed effects.

2.5.3.1 Estimation and Inference

Estimating LMMs parameters is more complex than estimating standard linear regression models due to the inclusion of random effects. Formally, the vector of observed outcomes $\mathbf{Y}$ can be represented as:

$$\mathbf{Y} \sim \mathcal{N}(\mathbf{X}\beta, \mathbf{V}(\lambda)),$$

In general, the outcome vector $\mathbf{Y}$ is assumed to follow a multivariate normal distribution with mean $\mathbf{X}\beta$ (determined by the fixed effects) and variance-covariance structure $\mathbf{V}(\lambda)$, which depends on a set of variance parameters λ capturing within and between-subject variability (Verbeke 2001).

The parameters of interest, both β and λ, are typically estimated via **Maximum Likelihood Estimation (MLE)** or **Restricted Maximum Likelihood (REML)**. The two main approaches are:

- **MLE:** MLE estimates both fixed and random effect parameters by maximizing the likelihood of the observed data.

$$\ell(\beta, \lambda) = -\tfrac{1}{2}\Big\{ |V(\lambda)| \; + \; (Y - X\beta)^T V(\lambda)^{-1}(Y - X\beta)\Big\}.$$

If λ were known, the MLE for β would be given by the **Generalized Least Squares (GLS) estimator**:

$$\hat{\beta}(\lambda) = \left(X^T V(\lambda)^{-1} X\right)^{-1} X^T V(\lambda)^{-1} Y,$$

which accounts for the covariance structure $V(\lambda)$. In practice, both β and λ are unknown. The estimation of λ is usually through iterative optimization techniques such as Fisher scoring (Pinheiro and Bates 2004a; Verbeke 2001).

- **REML:** REML addresses a key limitation of MLE, namely bias in estimating variance components. REML works by transforming the data into a new vector $U = A^T Y$, where A is orthogonal to the fixed-effects design matrix X. This transformation removes the influence of β when estimating λ (i.e., removing dependence on β), thereby reducing bias in variance component estimation. The transformed data then follow the distribution

$$U \sim \mathcal{N}(0, A^T V(\lambda) A),$$

  so that the likelihood depends only on λ allowing unbiased estimation of the variance components.

The REML log-likelihood is then:

$$\tilde{\ell}(\lambda) = -\tfrac{1}{2}\Big\{ |A^T V(\lambda) A| \; + \; U^T (V(\lambda)^{-1}) U\Big\}$$

Maximizing $\tilde{\ell}(\lambda)$ with respect to λ gives the REML estimates $\hat{\lambda}_{\text{REML}}$. The corresponding fixed-effect $\hat{\beta}_{\text{REML}}$ can be computed using the GLS framework:

$$\hat{\beta}_{\text{REML}} = \hat{\beta}(\hat{\lambda}_{\text{REML}}) = \left(X^T V(\hat{\lambda}_{\text{REML}})^{-1} X\right)^{-1} X^T V(\hat{\lambda}_{\text{REML}})^{-1} Y.$$

Importantly, the REML estimator $\hat{\lambda}_{\text{REML}}$ does not depend on the specific choice of transformation matrix A (Pinheiro and Bates 2004a; Verbeke 2001).

Inference: Inference for the linear mixed-effects model involves both fixed and random effects.

- *Fixed effects:* Inference is based on the estimated coefficients $\hat{\beta}$ and their covariance matrix

$$\text{Var}(\hat{\beta}) = \left(X^T V(\hat{\lambda})^{-1} X\right)^{-1},$$

and hypothesis tests and confidence intervals for fixed effects are derived from their estimated standard errors, which are adjusted to account for the hierarchical covariance structure (Verbeke 2001).

- *Random effects (variance components):* Inference is typically likelihood based, REML is preferred for estimating variance components due to reduced bias, while MLE is more appropriate for likelihood ratio tests in model comparisons (Verbeke 2001).

In this simulation study, all LMM models (both the gold standard pooled analysis and FSDA-based analysis) were estimated using **REML**, using the `lme4` R package version 1.1-37, via the `lmer()` function (Bates et al. 2015). This ensured unbiased estimation of the variance components while allowing valid inference on the fixed effects.

2.6 Simulation Study

This simulation study plan follows the ADEMP framework by Morris et al. (2019), which breaks down simulation studies into five key elements: **aims, data-generation mechanisms, estimands, methods, and performance measures**. The following subsections describe each element in detail.

2.6.1 Aims

The primary aim of this simulation study was to evaluate the performance of FSDA in estimating parameters of LMMs applied to multi-site longitudinal data. Specifically, the investigation sought to:

- **Assess accuracy**, by quantifying the bias of fixed-effects and variance component estimates relative to their true values.
- **Assess precision**, by examining the variability of estimates across simulation replicates.
- **Assess overall performance**, through the mean squared error (MSE), which jointly reflects bias and variance.
- **Robustness** of FSDA performance when model complexity increases (e.g., additional random effects) or when heterogeneity across sites is introduced.

Additionally, **privacy risk** was evaluated separately in the thesis using disclosure risk metrics to assess whether synthetic data generated under FSDA adequately protect confidentiality.

By comparing the estimates from synthetic data to the "true" parameter values used in the simulation, we aimed to evaluate whether FSDA can simultaneously preserve analytical validity in multi-site data settings.

2.6.2 Data-generation Mechanisms

2.6.2.1 Study Design

The simulation was designed to reflect a **multi-center longitudinal cohort** with repeated measures of subjects nested within sites. The structure was motivated by the **IDEFICS/I.Family cohort**, a large European child cohort initiated as the *Identification and prevention of dietry- and lifestyle-induced health effects in children and infants (IDEFICS)* study and later extended through the *I.Family* follow-up (Ahrens et al. 2016; Wolters et al. 2022).

The simulated dataset included multiple sites, each with repeated measures for subjects. Specifically:

- **Sites**: 8 federated sites, each with its own local dataset.
- **Subjects**: Each site contributed a varying number of subjects, introducing heterogeneity in sample sizes across sites.
- **Follow-up Visits**: Subjects were assessed at two time points: baseline (T0) and follow-up after six years (T3).
- **Hierarchy**: Observations (i) are nested within subjects (j), which are in turn nested within sites (k).

This structure mirrors longitudinal multi-center designs, where repeated measurements are clustered within subjects, and subjects are further clustered within sites.

2.6.2.2 Population and Baseline Covariates

Baseline covariates were drawn from prespecified distributions to reflect realistic characteristics of children at study entry. Table 2.1 summarizes the distributions used as inputs for simulation.

Table 2.1: *Baseline covariates used for simulation inputs*

Variable	Distribution	Parameters	Description
Age_{T0}	Normal	$\mu = 6, \sigma^2 = 4$	Baseline age in years
Sex	Bernoulli	$p = 0.5$	Male (1), Female (0)
$\text{Time Outdoors}_{\text{T0}}$	Normal	$\mu = 2, \sigma^2 = 4$	Hours spent outdoors per day
$\text{Time Outdoors}_{\text{T3}}$	Normal	$\mu = 2, \sigma^2 = 2.25$	Hours spent outdoors per day at follow-up

The variables were simulated as follows:

- `Age`: Simulated as a continuous variable at two time points: baseline (T0) and follow-up (T3). The baseline age, Age_{T0}, was drawn from a normal distribution:

$$\text{Age}_{T0} \sim \mathcal{N}(6, 4)$$

  Age at follow-up was calculated as $\text{Age}_{T3} = \text{Age}_{T0} + 6$. representing a 6 year time lapse between assessments.

- `Sex`: A binary variable, (0 = female, 1 = male), drawn from a Bernoulli distribution with equal probability:

$$\text{Sex} \sim \text{Bernoulli}(0.5)$$

- `Time Outdoors`: Modeled as a normally distributed continuous variable, with different variances at the two time points to reflect changes in variability over time:

$$\text{Time Outdoors}_{T0} \sim \mathcal{N}(2, 4), \quad \text{Time Outdoors}_{T3} \sim \mathcal{N}(2, 2.25)$$

2.6.2.3 Site-Level and Subject-Level Random Effects

To capture the hierarchical variation, **two levels of random intercepts** were introduced:

- **Site-level random effects:**

Random effects at the site level, denoted as u_{site_j}, were drawn from a normal distribution to reflect variation between sites (between-site heterogeneity):

$$u_{\text{site}_j} \sim \mathcal{N}(0, 3).$$

A total of 8 unique site identifiers (`Site_ID`) were generated to represent 8 distinct sites in the study.

- **Subject-level random effects (nested within sites):**

Subjects were nested within sites, and each subject was assigned a unique identifier (`Subject_ID`). Subject-specific random effects, denoted $u_{\text{subject}_{ij}}$, were sampled from a normal distribution to account for individual-level variation (variability across subjects):

$$u_{\text{subject}_{ij}} \sim \mathcal{N}(0, 2).$$

Each site was assigned a variable number of subjects (`nSubjects`) to simulate realistic heterogeneity in site sizes, ensuring a more natural data generation process reflective of multi-center study designs.

2.6.2.4 Simulation Outcome Model

The outcome, **Vitamin D level**, denoted by Y_{ijk}, was generated from LMMs that includes both fixed effects, random effects, and noise, based on the predefined coefficients inspired by the literature (Wolters et al. 2022). The model equation is as follows:

$$Y_{ijk} = \beta_0 + \beta_1 \cdot \text{Sex}_j + \beta_2 \cdot \text{Age}_{ij} + \beta_3 \cdot \text{Time Outdoors}_{ij} + u_{\text{subject}_{j[k]}} + u_{\text{site}_k} + \epsilon$$

where:

- Y_{ijk}: Vitamin D level for observation k of subject i in site j,
- $\beta_0, \beta_1, \beta_2, \beta_3$: fixed effect coefficients for the intercept, sex, age, and time outdoors,
- $u_{\text{subject}j[k]} \sim \mathcal{N}(0, \sigma^2_{\text{subject}})$: subject-specific random effect,
- $u_{\text{site}k} \sim \mathcal{N}(0, \sigma^2_{\text{site}})$: site-specific random effect,
- $\epsilon \sim \mathcal{N}(0, \sigma^2_\epsilon)$: residual error term.

For the simulation, the model was specified with the following parameter values:

$$Y_{ijk} = 45 + 2 \cdot \text{Sex}_j - 0.5 \cdot \text{Age}_{ij} + 1.5 \cdot \text{Time Outdoors}_{ij} + u_{\text{site}_k} + u_{\text{subject}_{j[k]}} + \epsilon,$$

with $\epsilon \sim \mathcal{N}(0, 14^2)$.

The parameter values used to specify this outcome model are summarized in Table 2.2. These values include fixed effects for sex, age, and time outdoors which were specified to induce realistic associations between covariates and outcome, as well as variance components for site-level and subject-level random effects to capture hierarchical structure. The residual variance was intentionally set to a relatively high value ($\epsilon \sim \mathcal{N}(0, 14^2)$) to reflect the high level of unexplained variability typically observed in real-world multi-site data, and allowing for a robust evaluation of FSDA under realistic data conditions. They serve as the "true" parameters against which the performance of the estimation methods were evaluated in the simulation study. Additionally, a binary indicator of Vitamin D deficiency was defined:

$$\text{Deficient}_{ijk} = \begin{cases} 1, & \text{if } Y_{ijk} \leq 50, \\ 0, & \text{if } Y_{ijk} > 50, \end{cases}$$

allowing us to evaluate both continuous outcome (Vitamin D concentration) and binary outcome (deficiency status).

Table 2.2: *Parameters of the outcome model used in the simulation*

Parameter	Value	Description
Intercept	$\beta_0 = 45$	Model intercept
Sex	$\beta_1 = 2$	Effect of being male (vs female)
Age	$\beta_2 = -0.5$	Effect of one-year increase in age
Time Outdoors	$\beta_3 = 1.5$	Effect of one-hour increase outdoors
Site random effect variance	$\sigma^2_{\text{site}} = 3^2$	Variance of site-level random intercepts
Subject random effect variance	$\sigma^2_{\text{subject}} = 2^2$	Variance of subject-level random intercepts
Residual variance	$\sigma^2_\epsilon = 14^2$	Residual variance of Vitamin D level

2.6.2.5 Synthetic Data Generation Process

The synthetic datasets were generated in line with the *sequential conditioning modeling* framework of the `synthpop` package (Nowok et al. 2016). In this approach, variables are synthesized one-by-one from condi-

tional distributions, such that each new variable is drawn conditional on those already synthesized (Nowok et al. 2016; Raab et al. 2017). This sequential process ensures that multivariate dependencies are preserved, thereby improving the utility of the resulting synthetic datasets (Raab et al. 2021; Snoke et al. 2018).

Several key guidelines as detailed by Raab et al. (2017) and Raab et al. (2021) were applied to tailor the synthesis process to the structure of our longitudinal and multi-site dataset:

1. **Order of Variables:** Variables were synthesized sequentially, with the order of synthesis determined by the `visit.sequence` parameter of `syn()`. When no explicit sequence is provided, as in our implementation, `synthpop` defaults to the *column order of the dataset*. This means that each variable is synthesized in turn, conditional on those already synthesized earlier in the dataset. Core variables such as `Site_ID`, `Sex`, `Age`, and `Time Outdoors` were synthesized before outcome variables (`Vitamine D level` and `Deficiency`). This ordering ensures that predictors are generated first and the outcomes are generated later, so the information flows in the correct direction. In this way, outcome variable is prevented from influencing the predictors and the natural relationship between variables is preserved (Nowok et al. 2016).

2. **Synthesis Method(s):** Synthesis methods were chosen per variable, in line with best practice:

- Continuous variables (`Age`, `Time Outdoors`, and `Vitamin D level`) were synthesized using CART, which is suited to capture complex dependencies and hierarchical structures in the data.
- Binary variables (`Sex`, `Deficient`) were synthesized with logistic regression.

This approach follows recommendations to match the synthesis method to the data type while ensuring utility in complex structure (Nowok et al. 2016; Raab et al. 2021).

3. **Stratified Synthesis:** To preserve site-specific distributions and heterogeneity, the data were stratified by `Site_ID` using the `syn.strata()` function. Stratification ensures that synthetic datasets reflect the variability across sites, a feature emphasized as critical for clustered or multi-center data (Nowok et al. 2016; Raab et al. 2021).

4. **Nested Synthesis for Hierarchical Identifiers:** For variables with a nested structure (e.g., `Subject_ID` nested within `Site_ID`), synthesis was performed using the `syn.nested()` function. This approach generates subject identifiers conditional on site membership, thereby maintaining both the hierarchical clustering of individuals within sites and the correlation of repeated measurements within subjects (Nowok et al. 2016; Raab et al. 2021).

5. **Predictor Matrix Adjustments:** The predictor matrix determines which variables are used to generate others. Identifiers such as `Site_ID` and `Subject_ID` were excluded as predictors to avoid overfitting and ensure that synthetic values are driven by meaningful covariates To preserve the hierarchical structure of the data, stratified synthesis was used to generate data within each site, and nested synthesis was

applied to generate `Subject_ID` within sites. For the synthesis of `Subject_ID` itself, was performed using cluster-level information (`Site_ID`, `u_site`, `nSubjects`, `u_subject`) to preserve within-site correlations. In this way, dependencies between observations are maintained without directly relying on identifiers (Nowok et al. 2016; Raab et al. 2021; Raab et al. 2017).

6. **Proper Synthesis:** All synthesis was conducted under *proper synthesis* (i.e., incorporating random draws from posterior predictive distributions), which is recommended to better reflect uncertainty in the synthetic process (Nowok et al. 2016).

The combination of these synthesis processes allowed us to preserve both the marginal distributions of key covariates and the correlation structure between variables. Following the published guidelines ensured that the synthetic datasets maintain utility. The privacy protection afforded by the synthesis process was subsequently evaluated using disclosure risk measures.

2.6.2.6 Utility Assessment of the Synthetic Data

The statistical utility of the synthetic data was evaluated using the **propensity score mean squared error (pMSE)** and its standardized form, the **standardized pMSE (S_pMSE)**, following the framework of Snoke et al. (2018) and Raab et al. (2021), and as implemented in the `synthpop` package (Nowok et al. 2016). For measuring utility, classification models are used to predict whether each record comes from the original or synthetic datasets. Commonly used classifiers include logistic regression (a parametric method) and non-parametric CART, which is more flexible for complex data (Raab et al. 2021; Snoke et al. 2018).

The pMSE quantifies how well the classifier distinguishes between original and synthetic data. Under the null hypothesis that synthetic data are indistinguishable from the original, the pMSE follows a known distribution (the *null distribution*). Specifically, the original data of size n_{orig} and synthetic data of size n_{syn} are combined into a single dataset of size $N = n_{\text{orig}} + n_{\text{syn}}$, with an indicator variable I denoting the data source ($I = 0$ for original, $I = 1$ for synthetic). A propensity score model is then fitted to predict the probability that each record is synthetic. Let $\hat{p}_r$ denote the predicted probability for record r, and define $c = \frac{n_{\text{syn}}}{N}$, as the true proportion of synthetic records in the combined dataset (Raab et al. 2021). The pMSE is given by

$$\text{pMSE} = \frac{1}{N} \sum_{r=1}^{N} (\hat{p}_r - c)^2.$$

A small pMSE indicates that the classifier cannot reliably separate original from synthetic records, implying that the two datasets are statistically similar with respect to the predictors used (Raab et al. 2021; Snoke et al. 2018).

However, raw pMSE values are not directly comparable across settings, as they depend on factors such as **sample size (N)**, the **proportion of synthetic observations** $c = n_2/N$, and the **complexity of the propensity score model** used to discriminate between datasets (Nowok et al. 2016; Snoke et al. 2018).

To account for this, the standardized pMSE (S_pMSE) was also computed, which standardizes the observed statistic relative to its **null distribution** (i.e., the distribution of pSME expected under **correct syn-**

thesis, where synthetic data are generated from the same underlying model as the original data). For logistic regression models, analytical results for the null expectation and standard deviation have been derived (Snoke et al. 2018):

$$\mathbb{E}_0[\text{pMSE}] = (k-1)\left(\frac{n_1}{N}\right)^2\left(\frac{n_2}{N}\right)\frac{1}{N} = \frac{(k-1)(1-c)^2\,c}{N},$$

$$SD_0[pMSE] = \sqrt{2(k-1)}\left(\frac{n_1}{N}\right)^2\left(\frac{n_2}{N}\right)\frac{1}{N} = \frac{\sqrt{2(k-1)}(1-c)^2\,c}{N}.$$

In the common case when $n_1 = n_2$, these simplify to:

$$\mathbb{E}_0[pMSE] = \frac{(k-1)}{8N}, \quad SD_0[pMSE] = \frac{\sqrt{2(k-1)}}{8N}$$

The standardized form is then given by:

$$S_{pMSE} = \frac{pMSE - \mathbb{E}_0[pMSE]}{SD_0[pMSE]}.$$

Under correct synthesis, $S_{pMSE} \approx 0$, while increasingly positive values indicate poorer similarity between the original and synthetic data. In practice, Raab et al. (2021) suggest that pMSE values below 10 can be regarded as indicative of adequate utility.

When the propensity model is non-parametric such as CART, no closed-form null distribution is available. In such cases, $\mathbb{E}_0[pMSE]$ and $SD_0[pMSE]$ are estimated empirically using a **paired resampling** approach: the original dataset is repeatedly split into two halves and treated as "original" and "synthetic", and a CART model is fitted to distinguish between them. The resulting pMSE values are averaged across repetitions to approximate the null distribution. This empirical approach allows S_pMSE to be calculated and interpreted even in the absence of analytical formulas (Nowok et al. 2016; Snoke et al. 2018).

In this thesis, for computing the utility, we used a CART-based propensity model to be able to accommodate mixed data types. The empirical approach proposed by Snoke et al. (2018) for estimating the null distribution of pMSE was computationally not feasible due to the high dimensionality of the dataset and the number of replications required. For this reason, resampling was set to `"none"` in `compare()` function, and pMSE values were computed directly without estimating the null expectation or standard deviation.

Because the null distribution was not estimated, S_pMSE could not be computed for the overall utility. However, S_pMSE could still be computed for individual variables using analytical approximations based on logistic models. Raw pMSE values were reported for overall utility, and variable-specific pMSE values were used to examine individual variable contribution to utility.

In this thesis, all utility assessment procedures were computed consistently across simulations to ensure comparability using the `utility.gen()` function in the synthpop R package, which by default uses CART to estimate propensity scores.

2.6.2.7 Disclosure Risk Assessment

Disclosure risk assessment is a critical step in evaluating the privacy protection of synthetic data. It examines the potential for intruders to either re-identify individuals (**identity disclosure**) or infer sensitive attributes (**attribute disclosure**). The framework and measures applied in this investigation follow the methodology of Raab et al. (2024), who extend the `synthpop` package in R to compute disclosure risk for synthetic data.

Notation and Formal Definitions:

Let q denote a **quasi-identifier** combination, i.e., a set of key variables that an intruder might plausibly know about an individual. Also, Q is the set of all distinct quasi-identifier groups, where each $q \in Q$ corresponds to one unique set of values across the chosen keys. In this thesis, the keys are:

- `Site_ID`, `Sex`, `Visit`, `Age`, and `Time Outdoors`.

Sensitive variables (which we ideally want to protect) considered potential **targets** are:

- `Subject_ID`, `Vitamin D level`, and `Deficient`.

To quantify disclosure risk, we define the following counts:

- $d_{.q}$: number of records in the original dataset with quasi-identifier q.
- d_{tq}: number of original records with keys q and target value t.
- $s_{.q}$: number of records in the synthetic dataset with q.
- s_{tq}: number of records in the synthetic with q and target value t.
- N_d: total number of records in the original dataset.
- N_s: total number of records in the synthetic dataset.
- N_d only: number of records in the original dataset that appear **only** in the original, i.e., their key combination (q) does not appear in the synthetic data.
- N_d both: number of records in the original dataset that also appear in the synthetic dataset, i.e., their key combination (q) exists in both datasets.
- N_s only: number of records in the synthetic dataset that appear **only** in the synthetic data, i.e., their key combination (q) does not exist in the original data.
- N_s both: number of records in the synthetic dataset that also exist in the original dataset, i.e., their key combination (q) exists in both datasets.

If a quasi-identifier appears in the original but not in the synthetic dataset, then $s_{.q} = s_{tq} = 0$. Conversely, if q is present in the synthetic but not in the original, then $d_{.q} = d_{tq} = 0$.

This alignment ensures that identity and attribute disclosure metrics can be computed consistently across both datasets, even when some quasi-identifiers are missing from one dataset.

Table 2.3: *Structure of quasi-identifier (q = Site_ID, Sex, Visit, Age, Time Outdoors) and target (t = Subject_ID, Vitamin D level, Deficient) counts in the original (upper) and synthetic (lower) datasets.*

	Only in Original	In Both	Only in Synthetic	Total
t = Subject_ID	d_{1q}	d_{1q}	0	$d_{1.}$
t = Vitamin_D_level	d_{2q}	d_{2q}	0	$d_{2.}$
t = Deficient	d_{3q}	d_{3q}	0	$d_{3.}$
Column sums	$d_{.q}$	$d_{.q}$	0	N_d
Totals	$N_{d\ only}$	$N_{d\ both}$	0	N_d
t = Subject_ID	0	s_{1q}	s_{1q}	$s_{1.}$
t = Vitamin_D_level	0	s_{2q}	s_{2q}	$s_{2.}$
t = Deficient	0	s_{3q}	s_{3q}	$s_{3.}$
Column sums	0	$s_{.q}$	$s_{.q}$	N_s
Totals	0	$N_{s\ both}$	$N_{s\ only}$	N_s

Note: The quasi-identifiers (q = Site_ID, Sex, Visit, Age, Time Outdoors) are cross-classified with the sensitive target variables (t Subject_ID, Vitamin D level, Deficient). The counts d_{tq} represent the number of original records for a given key-target combination, while s_{tq} are the corresponding counts from the synthetic dataset. The partition into "only in original", "in both", and "only in synthetic" provides the foundation for calculating disclosure measures such as Unique in Original (UiO), Replicated Uniques (RepU), and attribute disclosure (DiSCO). Adapted from: Raab et al. (2024).

Identity Disclosure Measures Identity disclosure arises when a record in the original dataset can be uniquely matched to a record in the synthetic dataset using quasi-identifiers (q) (Raab et al. 2024). In practice, this simulates an intruder who knows some characteristics of an individual and attempts to find them in the synthetic data.

However, when quasi-identifiers include **continuous variables**, exact matches between records in the original and synthetic datasets are rare (Raab et al. 2024). This can artificially inflate uniqueness in the original dataset (leading to $UiO \approx 100\%$) while driving the apparent re-identification risk in the synthetic dataset ($RepU$) close to zero.

To address this, **continuous variables were grouped (binned)** prior to the calculation of disclosure measures. Grouping ensures that uniqueness reflects realistic intruder knowledge, as intruders are unlikely to know an individual's characteristics with exact precision (Raab et al. 2024).

In this analysis, the continuous variables Age, Time Outdoors, and Vitamin D level were included either as quasi-identifiers (for identity disclosure) or as target variables (for attribute disclosure). These variables

were automatically grouped within the `multi.disclosure()` function of the synthpop R package, which creates categorical bands based on the distribution of each variable. This allows for consistent comparison of quasi-identifiers and target variables across both datasets while preventing artificial inflation of uniqueness.

Following the recommendations by Raab et al. (2024), three target variables (`Vitamin D level`, `Subject_ID`, `Deficient`) were selected for disclosure risk assessment. `Vitamin D level` represents sensitive health information that an intruder might attempt to infer, while `Deficient` (a binary indicator of vitamin D deficiency) reflects an important derived health status. In addition, the continuous quasi-identifiers (`Age` and `Time Outdoors`) were also grouped to prevent unrealistic exact matching across records. Both continuous keys and targets were automatically divided into **quantile-based categories** (default: five bins) by the `multi.disclosure()` function. This binning procedure ensures that identity disclosure measures and attribute disclosure measures are interpretable and reflect plausible levels of intruder knowledge rather than artefacts of continuous precision.

The following measures quantify the risk of identity disclosure:

1. **Unique in Original (UiO):** This represents the percentage of records in the original dataset that are unique based on the chosen quasi-identifiers.

$$\text{UiO} = 100 \times \frac{\sum_{q \in Q}(d_{.q} \mid d_{.q} = 1)}{N_d}.$$

   A higher UiO means more individuals in the real data can be uniquely identified using these keys. This provides a baseline level of uniqueness risk inherent to the original dataset.

2. **Unique in Synthetic (UiS):** This is the percentage of records that are unique in the synthetic data, but only among those that were unique in the original.

$$\text{UiS} = 100 \times \frac{\sum_{q \in Q}(s_{.q} \mid d_{.q} = 1)}{N_d}.$$

It shows whether unique original records also tend to appear as unique in the synthetic dataset. A low UiS indicates the synthesis process has reduced uniqueness, lowering disclosure risk.

3. **Unique in Original and Synthetic (UiOiS):** This measure captures the percentage of original unique records that are present in the synthetic dataset but not necessarily unique.

$$\text{UiOiS} = 100 \times \frac{\sum_{q \in Q}(d_{.q} = 1 \mid s_{.q} = 1 \wedge d_{.q} > 0)}{N_d}.$$

From an intruder's perspective, these are records that could potentially be re-identified with high confidence.

4. **Replicated Uniques (RepU):** It quantifies the proportion of synthetic records that are unique in the original data, and reproduced as unique in the synthetic data.

$$\text{RepU} = 100 \times \frac{\sum_{q \in Q}(s_{.q} \mid d_{.q} = 1 \wedge s_{.q} = 1)}{N_d}.$$

These cases represents the *highest re-identification risk*, because an intruder who knows the keys can find a one-to-one match in the synthetic dataset.

Recommended measure: Following Raab et al. (2024), RepU is the preferred indicator of identity disclosure risk, since it quantifies the proportion of unique records in the original that are also unique in the synthetic.

Attribute Disclosure Measures Attribute disclosure occurs when an intruder, who already knows a set of quasi-identifiers (q), can infer the value of a sensitive target variable (t). In other words, disclosure happens if in a group defined by q, all records have the same target value t. To formalize this, we describe the following proportions defined by Raab et al. (2024):

- $pd_{tq} = d_{tq}/d_{.q}$ - the proportion of records in the original data with keys q that take target values t.
- $ps_{tq} = s_{tq}/s_{.q}$ - the proportion of records in the synthetic data with keys q that take target values t.

These proportions allow us to determine whether the target value is predictable from the keys.

The following measures quantify the risk:

1. **Disclosive in Original (Dorig):** This is the percentage of records in the original data where the target variable t is predictable from the quasi-identifiers q. In other words, if within a group defined by keys q, all records share the same target value t, then every record in that group is counted as disclosive in the original data.

$$D_{orig} = 100 \times \frac{\sum^{q}\sum^{t}(d_{tq} \,|\, pd_{tq} = 1)}{N_d}.$$

2. **Disclosive in Synthetic (Dsyn):** This is the percentage of records in the synthetic dataset where the target variable t is predictable from the quasi-identifiers q. That is, in the synthetic data, if every record in a group defined by keys q has the same target value t, then those records are considered disclosive in the synthetic data.

$$D_{syn} = 100 \times \frac{\sum^{q}\sum^{t}(s_{tq} \,|\, ps_{tq} = 1)}{N_s}.$$

3. **In Synthetic Original (iSO):** This measure captures the proportion of records in the original dataset

for which the quasi-identifier combination q also appears in the synthetic dataset.

$$iSO = 100 \times \frac{\sum^{q}\sum^{t}(d_{tq} \,|\, s_{tq} > 0)}{N_d}$$

Intuitively, iSO measures how many original records are at risk of disclosure simply because their key profile is reproducible in the synthetic data. However, at this stage no information about the target variable is yet revealed.

4. **Disclosive in Synthetic Correct Original (DiSCO):** This is the percentage of original records for which q appears in the synthetic data with a single target value t, and that value matches the original. It reflects cases where the synthetic data allow correct inference of a sensitive attribute.

$$\text{DiSCO} = 100 \times \frac{\sum^{q}\sum^{t}(d_{tq} \,|\, ps_{tq} = 1)}{N_d}$$

Intuitively, this measure asks: If an intruder looks up a key combination q in the synthetic data and sees that it always maps to a single value of t, how often would that inference also be correct in the original data?

5. **Disclosive in Synthetic and Original (DiSDiO):** This is a stricter version of DiSCO. It only counts records that are fully disclosive in both the original and the synthetic data. While more conservative, it is less informative from the intruder's perspective, since they would not know whether a record is disclosive in the original.

$$\text{DiSDiO} = 100 \times \frac{\sum^{q}\sum^{t}(d_{tq} \,|\, ps_{tq} = 1 \wedge pd_{tq} = 1)}{N_d}$$

Recommended measure: According to Raab et al. (2024), DiSCO is the preferred attribute disclosure measure, since it best represents what an intruder could actually infer from synthetic data alone.

Following Raab et al. (2024), our investigation adopts **RepU** as the primary identity disclosure risk measure and **DiSCO** as the primary attribute disclosure risk measure. These metrics directly capture the extent to which synthetic data could compromise privacy while maintaining analytical utility. Disclosure measures were implemented in R using the `multi.disclosure()` function from *synthpop*, which computes UiO, RepU, Dorig, and DiSCO across multiple target variables simultaneously.

2.6.3 Estimands

The estimands of interest in this simulation study are the population average effects of baseline covariates on Vitamin D levels, together with the variance components arising from the hierarchical data structure. The

target population corresponds to children and adolescents simulated to resemble the analysis from Wolters et al. (2022) based on the IDEFICS/I.Family cohort. The outcome of interest is a continuous measure of Vitamin D levels, repeatedly observed for each subject.

The data-generating process was based on a LMM:

$$Y_{ijk} = \beta_0 + \beta_1 \cdot \text{Sex}_j + \beta_2 \cdot \text{Age}_{ij} + \beta_3 \cdot \text{Time Outdoors}_{ij} + u_{\text{subject}_{j[k]}} + u_{\text{site}_k} + \epsilon,$$

where Y_{ijk} denotes Vitamin D level for observation i of subject j in site k.

The estimands are as follows:

- *Fixed Effects (population average parameters):*
  - β_0: Mean baseline Vitamin D level in the population.
  - β_1: Average difference in Vitamin D levels between males and females.
  - β_2: Average change in Vitamin D level per additional year of age.
  - β_3: Average change in Vitamin D level per extra hour spent outdoors.
- *Random Effects (variance components):*
  - σ^2_{site}: Between-site variance in Vitamin D levels, capturing heterogeneity across study sites.
  - $\sigma^2_{\text{subject}}$: Between-subject variance within sites, accounting for correlation across repeated measures.
  - σ^2: Residual variance unexplained by fixed effects and random effects.

The true values of all parameters were fixed during the data-generation process and serve as the benchmark against which estimator performance is evaluated. These estimands provide the target quantities for assessing whether FSDA can preserve valid statistical inference relative to pooled analysis of the original data.

2.6.4 Methods under Evaluation

We considered a single simulation setting in which the same data-generating mechanism was applied for **1000 simulation replicates** (chosen to balance precision with computational feasibility) (Morris et al. 2019). Each replicate consisted of a multi-site hierarchical dataset with subject-level and site-level random effects. For each replicate, the data were analyzed using two approaches:

1. **Gold Standard Pooled Analysis**: In the pooled analysis, all individual-level data from multiple sites were centrally combined, allowing for direct modeling using LMM. This method does not impose any privacy constraints and is considered the benchmark for comparison. The model was fitted using the pooled dataset, capturing both subject and site-level variation through random intercepts, while adjusting for key covariates as (Sex, Age, Time Outdoors) as fixed effects.

2. **FSDA**: As described in detail in Section 2.3, for each site we independently generated a **synthetic dataset** based on their original data using the `synthpop` package. These synthetic datasets were then centrally pooled and analyzed with the same LMM (via REML) as in the gold standard analysis.

In both approaches, all parameters were estimated using **Restricted Maximum Likelihood (REML)** with the `lmer()` function from the lme4 package in R (Bates et al. 2015).

Model Comparison Workflow: The following procedure was followed for each simulation repetition:

1. Generate the hierarchical dataset according to the model in Section 2.6.2.
2. Analysis paths:
    - **Original pooled analysis:** Directly pool the simulated datasets across all sites.
    - **FSDA analysis:** At each site, synthesize the local dataset using the `synthpop` package, then centrally pool all synthetic datasets.
3. Fit the same linear mixed-effects model to:
    - The **original pooled data** (gold standard), and
    - The **pooled synthetic data** (FSDA).
4. Extract all estimands from both models.
5. Repeat steps 1-4 a **total of 1000 times**.

The goal of this comparison was to assess how closely FSDA approximates the inference obtained from the gold standard approach while preserving privacy.

2.6.5 Model Fitting and Software

The simulation study was implemented in R (4.4.2). Data generation was performed using the `simstudy` package (version 0.8.1), which provides a flexible framework for defining baseline covariates, specifying hierarchical data structures, and generating repeated measures.

The synthesis of federated datasets was conducted with the `synthpop` package (version 1.9-0), following best practices guidelines for stratification and nested synthesis to preserve clustering at the site and subject levels (Nowok et al. 2016; Raab et al. 2017). Synthetic datasets were generated under *proper synthesis*, ensuring that posterior predictive variability was incorporated to avoid overfitting and to reflect uncertainty in the synthesis process.

Mixed-effects models were fitted using the `lme4` package (version 1.1-36) (Bates et al. 2015). Estimation was performed via *restricted maximum likelihood (REML)* which is widely recommended for linear mixed

models as it produces less biased variance component estimates compared to maximum likelihood (ML), particularly in finite samples (Laird and Ware 1982).

To ensure computational efficiency, *parallelization* was implemented using the `purrr` (version 1.1.0) and `furrr` (version 0.3.1) packages, enabling simultaneous synthesis and model fitting across repetitions. Simulation repetitions were organized through *batch processing*, with intermediate results and errors recorded via a *logging framework*, ensuring reproducibility and fault tolerance.

Utility and disclosure risk assessment of the synthetic datasets was carried out using functions from the `synthpop` package. General utility was quantified using the propensity score mean squared error (pMSE) and its standardized form (S_pMSE), as implemented in `utility.gen()`, which assess whether synthetic datasets are statistically indistinguishable from the original data (Raab et al. 2021). Disclosure risk was evaluated via measure of identity disclosure based on quasi-indentifiers, comparing the overlap between original and synthetic records.

The simulation study was repeated *1000 times*, and results were aggregated to compute performance metrics including bias, variance, and mean squared error (MSE) statistics. These metrics were used to evaluate the performance of FSDA relative to pooled analyses of the original data.

2.6.6 Performance Measures

To evaluate whether FSDA preserves valid inference relative to the gold standard (pooled analyses of the original data), simulation-based performance measures were calculated across 1000 replicates. These measures directly reflect accuracy, precision, and overall performance.

In each simulation replicate, both FSDA and pooled analysis were used to estimate the fixed and random effects defined in Section 2.6.3 (*Estimands*). For each parameter, the following measures were computed:

Bias:

Bias measures the average difference or deviation of the estimated parameters $\hat{\theta}$ (fixed effects and variance components) from their true parameter values θ. Bias quantifies the accuracy of FSDA-based estimates. An unbiased estimator (biase of zero) is desirable.

$$\text{Bias}(\hat{\theta}) = \frac{1}{n_{sim}} \sum_{i=1}^{n_{sim}} \hat{\theta}_i - \theta$$

A small bias indicates that the estimation is on average, centered around the true value.

Relative Bias:

Because the parameters in this thesis were different in magnitude and units, bias was also expressed in relative terms to allow for easy comparison across all parameters. Relative bias scales the bias by the size of the true value and is represented as a percentage here.

$$\text{Relative Bias}(\hat{\theta}) = \frac{\text{Bias}(\hat{\theta})}{\theta} \times 100\%$$

A relative bias that is close to zero percent indicates that the estimator accurately recovers the true value, but large positive or negative values indicate overestimation and underestimation respectively.

Variance:

The variability of parameter estimates across simulation replicates. Variance reflects the precision of FSDA relative to pooled analyses of the original data.

$$\text{Var}(\hat{\theta}) = \frac{1}{n_{sim} - 1} \sum_{i=1}^{n_{sim}} \left(\hat{\theta}_i - \bar{\hat{\theta}}\right)^2$$

where $(\bar{\hat{\theta}})$ is the average estimate across simulations.

Mean Squared Error (MSE):

Defined as the sum of squared bias and variance, MSE provides a combined measure of overall estimator performance.

$$\text{MSE}(\hat{\theta}) = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta})$$

These metrics were summarized (mean and variability) across replicates for each fixed and random effect parameter to assess the performance of FSDA relative to pooled analyses of the original data.

In addition, **general utility** (statistical similarity between synthetic and original data) and **disclosure** risks (privacy protection) were evaluated separately, as described in Section 2.6.2.6 and Section 2.6.2.7, respectively.

3 Results

3.1 Overview

In this chapter, we present the results of this thesis, which is structured around four components.

First, we describe the **characteristics of the simulated and synthetic datasets**, summarising key covariates, outcome distributions, and clustering patterns across sites to provide context for subsequent analyses.

Second, we evaluate the **utility of the synthetic data** relative to the original datasets. This includes both overall utility assessment using established propensity score-based metrics and targeted evaluation of strategies for tuning the synthesis process.

Third, we report findings from the **simulation study**, which was designed to assess feasibility and validity of applying FSDA to LMMs in a multi-site longitudinal setting. Simulation results focus on bias, variance, and MSE of parameter estimates, as well as the robustness of FSDA under increased model complexity and heterogeneity.

Finally, we present the **disclosure risk analysis**, quantifying the extent to which synthetic datasets generated under FSDA protect confidentiality. This assessment is based on identity disclosure and attribute disclosure metrics, providing a complementary perspective to utility-focused evaluations.

Together, these results provide an integrated view of the trade-off between **analytical validity** and **privacy protection**, thereby addressing the aims and objectives of this thesis.

3.2 Descriptive Statistics of the Simulated Data

This subsection reports the realized characteristics of the simulated datasets. The population-level distributions and baseline covariates used in the data-generating mechanisms are summarized in Table 2.1, while Table 3.1 presents the descriptive statistics of the simulated dataset used in the analysis.

A total of approximately 80,000 subjects were generated across 8 sites, with each subject contributing up to two repeated measures (baseline at T0 and 6-year follow-up at T3). At baseline (T0), `Age` had a mean of 6 years (SD = 2), and `Sex` was balanced with roughly equal proportions of males and females. `Time Outdoors` at baseline had a mean of 2 hours/day (SD = 2). At follow-up (T3), `Age` was defined to increase by 6 years, resulting in a mean of 12 years (SD = 2). `Time Outdoors` at follow-up retained a mean of 2 hours/day but with reduced variability (SD = 1.5). The simulated outcome, `Vitamin D level`, had a mean of 47.0 nmol/L (SD = 14.5) at baseline and 44.1 nmol/L (SD = 14.4) at follow-up. Across both time points, about 60% of observations had Vitamin D levels below or equal to 50 nmol/L.

In addition to the fixed covariate distributions, variability was introduced through random effects. A site-level random intercept variance of $\sigma^2_{site} = 3^2$ and subject-level random intercept variance to $\sigma^2_{subject} = 2^2$ were specified, both assumed to be constant across all sites. The residual error variance was set to $\sigma^2_{\epsilon} = 14^2$.

Table 3.1: *Descriptive characteristics of the simulated dataset ($n \approx 80{,}000$ subjects across 8 sites). Continuous variables are summarized with means, standard deviations (SD), and ranges; categorical variables are summarized with proportions.*

Continuous variables	**Mean**	**SD**	**Min**	**Max**
Age_{T0}	5.99	2.00	-2.38	14.1
Age_{T3}	12.00	2.00	3.62	20.1
$\text{Time Outdoors}_{\text{T0}}$	1.99	2.00	-6.67	10.8
$\text{Time Outdoors}_{\text{T3}}$	2.01	1.49	-4.75	8.23
$\text{Vitamin D level}_{\text{T0}}$	47.0	14.5	-11.4	104.
$\text{Vitamin D level}_{\text{T3}}$	44.1	14.4	-16.6	105.
Categorical variables				
Sex (Male)		50.1%		
Deficient ($Y \leq 50$)		60.4%		

Age, Time Outdoors, and Vitamin D level were simulated from normal distributions, which resulted in small proportion of negative values (Age: 0.16% at T0; 0% at T3; Time Outdoors: 16.0% at T0; 8.9% at T3; Vitamin D level: 0.06% at T0; 0.10% at T3), which were retained for analysis.

These components induced clustering of Vitamin D levels within sites and subjects, consistent with the hierarchical structure of the linear mixed-effects model described in Section 2.6.3.

Figure 3.1 displays the distribution of Vitamin D levels across the eight simulated sites, illustrating variation between sites due to the specified site-level random effects. Similarly, Figure 3.2 shows histograms and density plots of the baseline covariates (Age, Time Outdoors, Sex, and Deficiency status), which reflect the distributions generated under the simulation design.

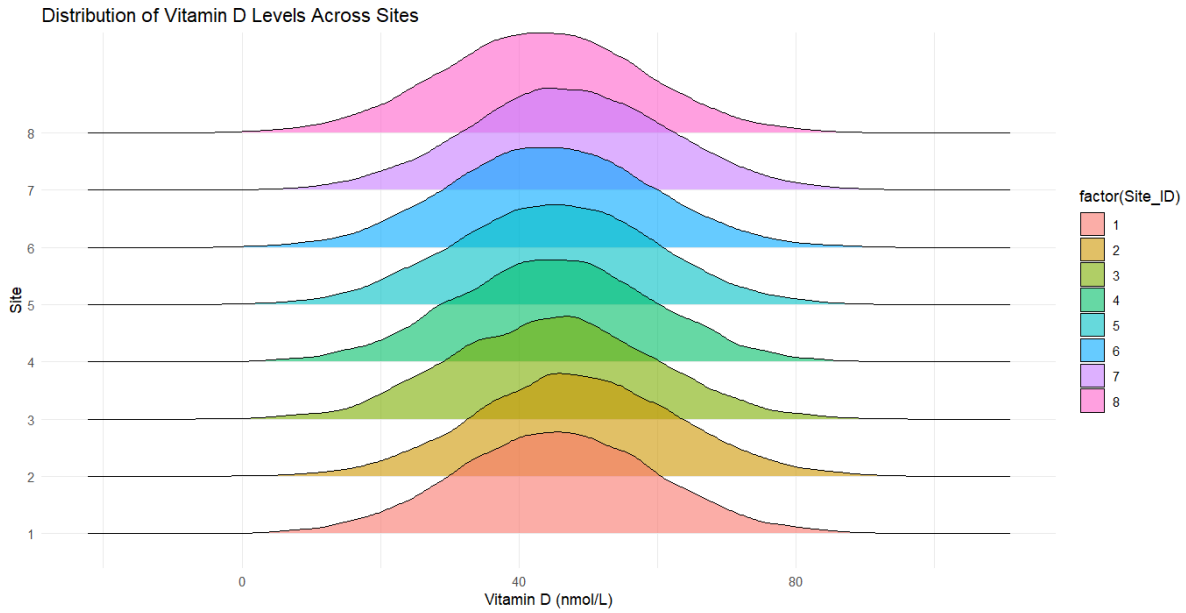


Figure 3.1: *Distribution of Vitamin D levels across the 8 simulated sites. Each ridge represents the density for one site, illustrating site-specific shifts in the outcome distribution. Although the distributions appear similar overall, the clustering reflects the imposed site-level random intercept variance ($\sigma^2_{site} = 3^2$), which induces variability in Vitamin D levels between sites.*

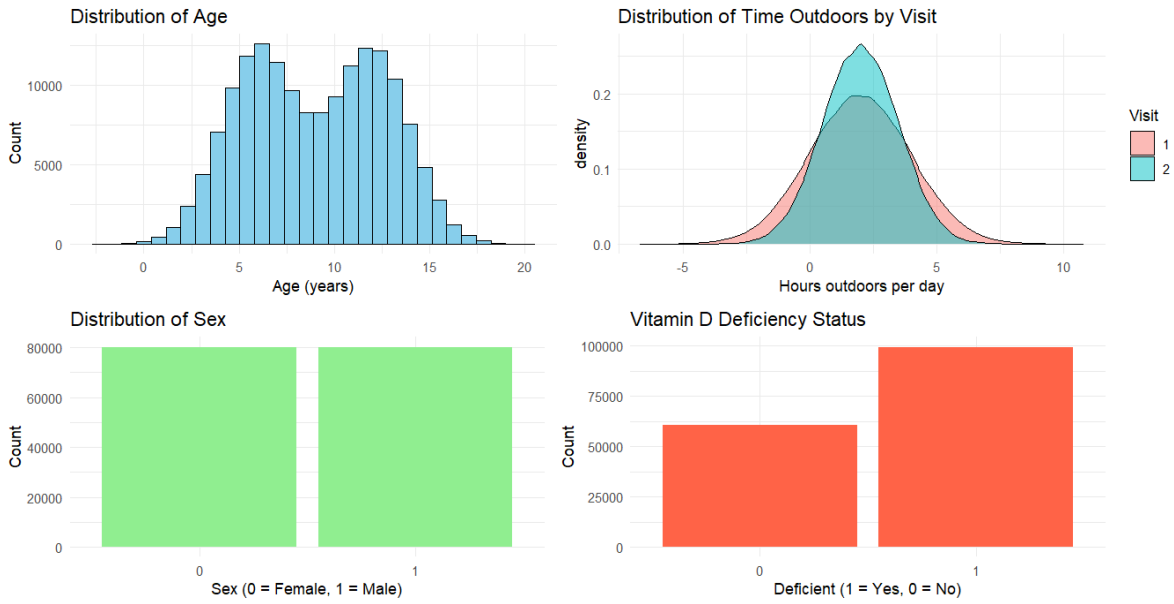


Figure 3.2: *Distributions of baseline covariates in the simulated dataset. (A) Age across all time points, showing bimodal distribution due to measurement at baseline ($T0$) and follow-up ($T3$). (B) Daily hours spent outdoors at baseline ($T0$) and follow-up ($T3$). (C) Sex distribution (0 = female, 1 = male). (D) Vitamin D deficiiency status, defined as $Y \leq 50$ nmol/L.*

Overall, the simulated datasets successfully reflected the predefined population structure and variability across sites, reproducing realistic patterns in age, sex distribution, time outdoors, and Vitamin D levels. The inclusion of site and subject-level random effects introduced meaningful clustering consistent with the hierarchical design of the study. These characteristics confirm that the data-generating process yielded a robust foundation for evaluating the performance of FSDA.

In the next subsection, we present the characteristics of the synthetic datasets generated from these simulated datasets. This provides an initial assessment of whether the synthetic process was able to reproduce the statistical properties and hierarchical structure observed in the original simulated data.

The full R code used to generate the simulated (original) data is provided in Appendix A.1 (see Section 2.6.5).

3.3 Characteristics of the Synthetic Data

The federated synthesis procedure (see Section 2.6.2.5) produced fully synthetic datasets in which all covariates and outcomes were replaced with draws from fitted conditional models. Proper synthesis was applied throughout, ensuring that random draws from the posterior predictive distributions were incorporated to reflect uncertainty.

Stratification by `Site_ID` and nested generation of `Subject_ID` were applied to preserve the hierarchical multi-site, multi-subject data structure. Variable-specific synthesis methods (CART for continuous variables,

logistic regression for binary outcome) were specified according to variable type, thereby reproducing the repeated measures design of the study.

Descriptive summaries of the synthetic datasets (Table 3.2) show the marginal distributions of Age, Time Outdoors, Sex, and Vitamin D levels. Site-level stratification and nested identifiers ensured that clustering of individuals within sites was retained.

Table 3.2: *Descriptive characteristics of the synthesized dataset ($n \approx 80{,}000$ subjects across 8 sites). Continuous variables are summarized with means, standard deviations (SD), and ranges; categorical variables are summarized with proportions.*

Continuous variables	**Mean**	**SD**	**Min**	**Max**
Age_{T0}	6.01	1.99	-2.12	13.5
Age_{T3}	12.0	2.00	3.62	19.9
$\text{Time Outdoors}_{\text{T0}}$	2.01	1.93	-6.67	10.4
$\text{Time Outdoors}_{\text{T3}}$	2.01	1.57	-5.79	9.47
$\text{Vitamin D level}_{\text{T0}}$	46.9	14.5	-11.6	102.
$\text{Vitamin D level}_{\text{T3}}$	44.0	14.4	-11.6	105.
Categorical variables				
Sex (Male)		50.5%		
Deficient ($Y \leq 50$)		62.2%		

Age, Time Outdoors, and Vitamin D level were synthesized based on the original distributions, which led to a small proportion of negative values (Age: 0.16% at T0; 0% at T3; Time Outdoors: 14.7% at T0; 9.6% at T3; Vitamin D level: 0.08% at T0; 0.11% at T3). These values were retained in the synthesized dataset to preserve the overall statistical properties of the original data.

The descriptive characteristics of the synthetic datasets indicated that the key marginal distributions of `Age`, `Time Outdoors`, `Sex`, and `Vitamin D levels` were closely reproduced through the synthesis process. Although these summaries confirm that the overall distributional properties were well maintained, they do not capture the extent to which site-level or subject-level structure were preserved. To address this, a more formal evaluation of **utility** was undertaken. Specifically, propensity score-based diagnostics (see Section 2.6.2.6) were applied to quantify how well the multivariate relationships in the synthetic data reflected those in the original dataset. This allows for a more rigorous assessment of whether the synthetic data retain the joint dependencies and hierarchical patterns essential for valid downstream analyses.

The full R code used to generate the synthetic data is provided in Appendix A.2.

3.4 Utility of the Synthesized Dataset

The statistical utility of the synthesized dataset was determined using the (pMSE) and its standardized form (S_pMSE), following the framework proposed by Snoke et al. (2018) and Raab et al. (2021), and implemented in the `synthpop` R package (Nowok et al. 2016). As described in Section 2.6.2.6, these measures evaluate how well a classification model can distinguish between the original and synthetic datasets. Lower pMSE and S_pMSE values indicate higher similarity between the original and synthetic datasets and therefore higher statistical utility.

Two levels of utility measures were computed:

- **Variable-specific utility scores**, obtained using the `compare()` function from the `synthpop` package, reporting both the raw pMSE and, where computable, the standardized form (S_pMSE) for each variable.
- **Overall utility score**, computed using the `utility.gen()` function, quantifying how well a classifier can distinguish synthetic from original records based on their joint distributions (i.e., quantifies how similar the multivariate relationships in the synthetic data are to those in the original data).

3.4.1 Variable-Specific Utility

Table 3.3 shows the variable-specific pMSE and S_pMSE values. The raw pMSE and S_pMSE values were generally small indicating that the synthetic data retained a high degree of similarity to the original data across most variables. Visual comparisons in Figures 3.3, 3.4, and 3.5 also show that the synthetic data closely reproduced the original marginal distributions for all key variables.

Table 3.3: *Variable-specific and overall utility measures comparing synthetic and original datasets, calculated using CART propensity score models. Low pMSE and S_pMSE indicate high similarity between synthetic and original data.*

Variable	pMSE	S_pMSE	df
Site_ID	0.000000	0.201173	3
u_site	0.058939	37720.98	4
nSubjects	0.000000	0.280008	3
Subject_ID	0.000007	4.287210	4
u_subject	0.000002	1.171947	4
Sex	0.000000	0.855625	1
Visit	0.000000	0.883601	1
Age	0.000005	3.366109	4
Time_Outdoors	0.000005	3.106851	4
Vitamin_D_level	0.000006	3.572126	4
Deficient	0.000000	0.182979	1
Overall (utility.gen)	0.216659	NA	–

Note: The inflated S_pMSE for `u_site` reflects the small number of site categories (8) and is not indicative of poor utility. Overall, pMSE values indicate that the synthetic data closely match the original data. df = degree of freedom.

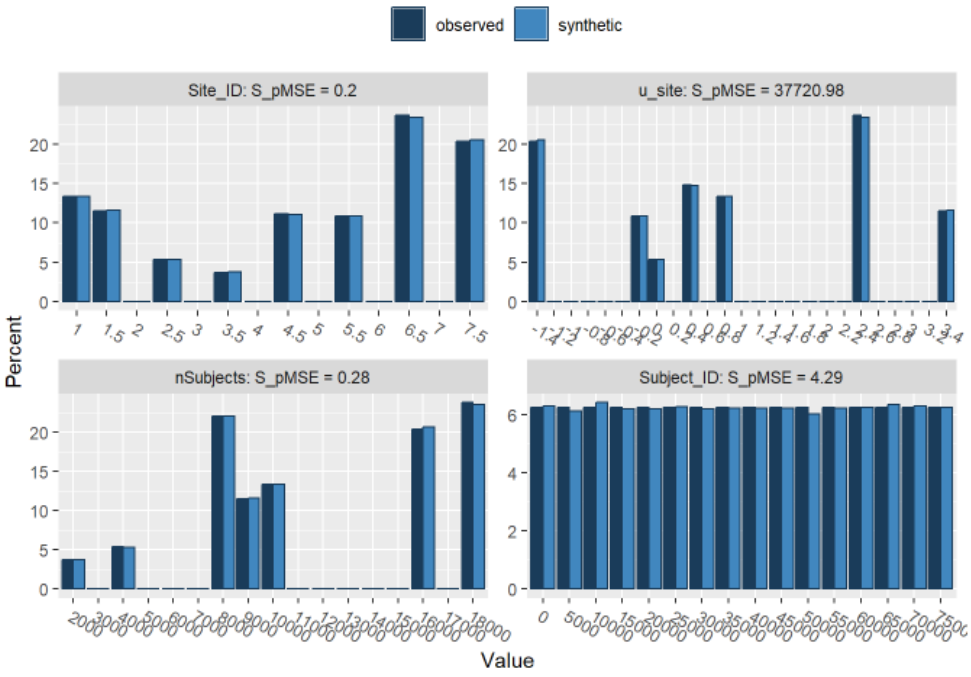


Figure 3.3: *Comparison of distributions for variables in (Panel 1) between synthetic and original datasets.*

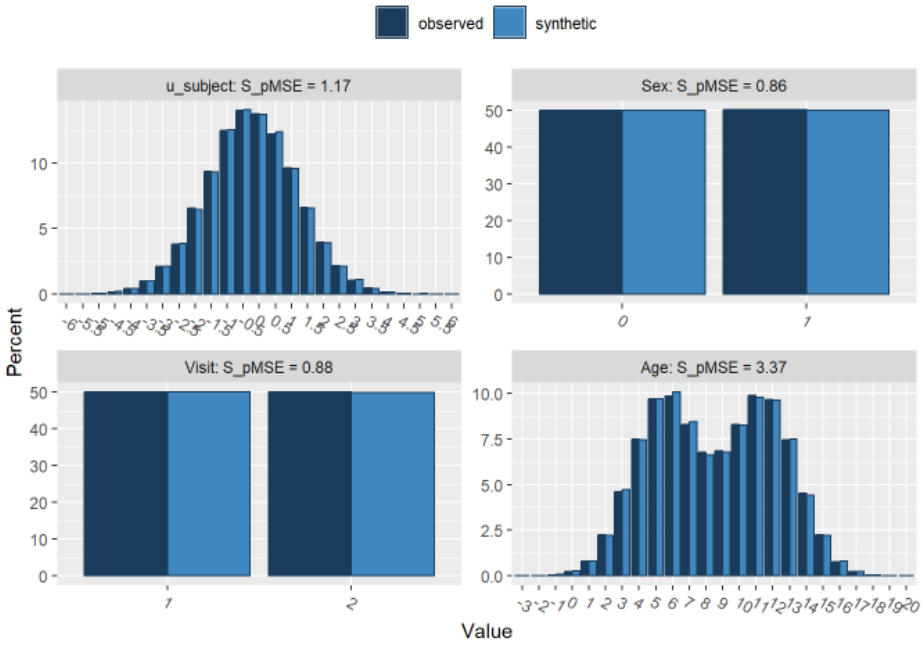


Figure 3.4: *Comparison of distributions for variables in (Panel 2) between synthetic and original datasets.*

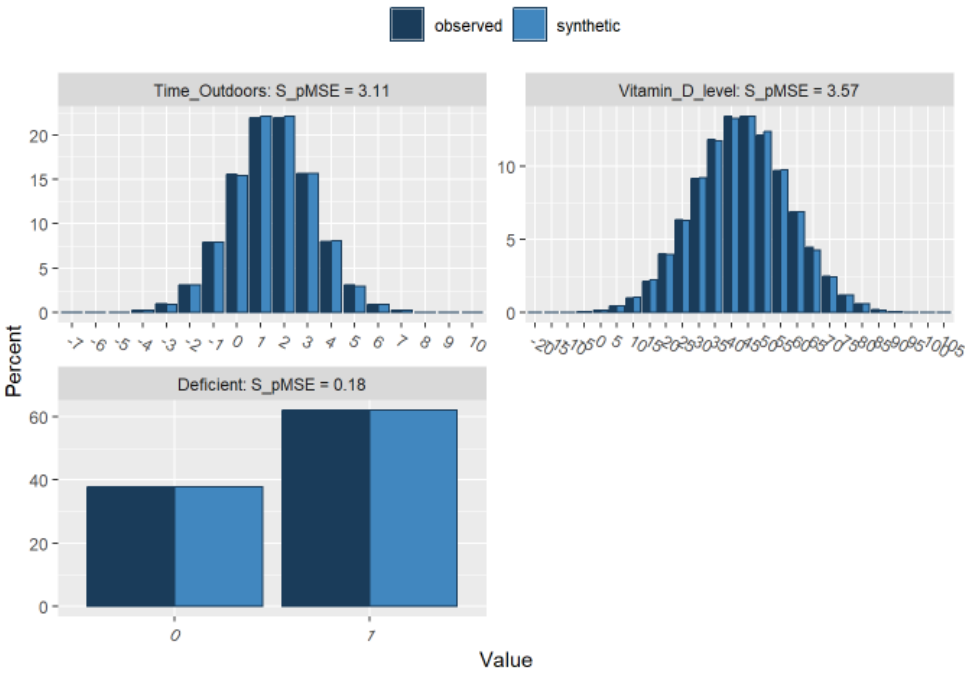


Figure 3.5: *Comparison of distributions for variables in (Panel 3) between synthetic and original datasets.*

For the site-level variable (`u_site`), the raw pMSE value remained small (0.0589), while the standardized S_pMSE value was large (37,721). This pattern reflects the small number of site categories rather than any substantive discrepancy in the synthetic data. Visual inspection of Figure 3.3 shows that the synthetic site-level variance (`u_site`) values closely resemble the original data, confirming that the synthesis captured the site-level variability very well. The subject_level random effect (`u_subject`) also showed good agreement between synthetic and original distributions. The utility measures computed for identifiers such as `Site_ID` and `Subject_ID` are not meaningful as these are simply labels. The primary focus of the utility assessment is therefore on variables that carry significant information, like covariates, outcomes and random effects.

3.4.2 Overall Utility

The overall raw pMSE value was 0.2167, indicating that the classification model (CART) was unable to distinguish between synthetic and original records under the joint distributional comparison i.e., the synthetic data are statistically similar to the original dataset. Standardized S_pMSE could not be computed because the resampling was set to “none”, therefore the empirical null distribution was not estimated. Raw pMSE was therefore used as the primary measure of overall utility.

Across the evaluated variables and the whole dataset, the synthetic data showed high statistical similarity to the original data. Variable-level pMSE and S_pMSE were small for nearly all variables, and the full

dataset raw pMSE showed strong overall utility. Minor anomalies in S_pMSE for hierarchical variables are consistent with their structural characteristics rather than indicative of poor synthesis.

Having established that the synthetic data closely reproduced the original data joint distribution, the next subsection presents the results of the simulation study comparing FSDA estimates with those from the gold standard pooled analysis of the original data in recovering the fixed effects and random effects.

The full R code for this utility assessment is provided in Appendix A.3 (see Section 2.6.5).

3.5 FSDA Simulation Study: Parameter Estimates

We evaluated the ability of FSDA to recover parameters from the prespecified LMM across 1,000 simulation repetitions (see Section 2.6.2 and 2.6.5 for the data-generating mechanism and software implementation, respectively). Results are presented in terms of descriptive statistics, performance metrics, and direct comparisons between FSDA synthetic data and the gold standard pooled analysis.

3.5.1 Gold Standard: Pooled Analysis

To establish a benchmark for comparison, the simulated datasets were first analyzed using a pooled analysis across all sites. Results were summarized over 1,000 repetitions, and detailed descriptive statistics are provided in Table 3.4.

Table 3.4: *Descriptive statistics for fixed and random effects under the gold standard pooled analysis (1,000 simulation repetitions). Mean, standard deviation (SD), and variance are reported.*

Parameter	Mean	SD	Variance
Fixed Effects			
Intercept	45.084	0.611	0.374
Sex	1.927	0.000	0.000
Age	-0.488	0.000	0.000
Time Outdoors	1.480	0.000	0.000
Random Effects			
Subject-level variance	1.503	0.003	0.000
Site-level variance	2.980	1.554	2.420
Residual			
Residual variance	196.641	0.001	0.000

To evaluate estimation accuracy, recovered estimates were compared to the true generating values defined in the simulation model (Section 2.6.2.4). Table 3.5 presents the true parameter values alongside the mean estimates, bias, relative bias, variance, and MSE for each parameter.

For the fixed effects, parameter estimates were recovered with minimal bias and very low variability across repetitions. Their computed relative bias were also very small, confirming high estimation accuracy. The `Intercept` had a mean 45.08 (bias = 0.084; relative bias = 0.19%; MSE = 0.381) with small variance

Table 3.5: *True values, mean estimates, bias, relative bias, variance, and mean squared error (MSE) for fixed and random effects under the gold standard pooled analysis.*

Parameter	True Value	Mean Estimate	Bias	Rel. Bias (%)	Variance	MSE
Fixed Effects						
Intercept	45.0	45.084	0.084	0.19	0.374	0.381
Sex	2.0	1.927	-0.073	-3.66	0.000	0.005
Age	-0.5	-0.488	0.012	-2.39	0.000	0.000
Time Outdoors	1.5	1.480	-0.021	-1.40	0.000	0.000
Random Effects						
Subject-level variance	2.0	1.503	-0.497	-24.86	0.000	0.247
Site-level variance	3.0	2.980	-0.020	-0.68	2.415	2.415
Residual						
Residual variance	196.0	196.641	0.641	0.33	0.000	0.411

(0.374), while covariate effects (`Sex`, `Age`, and `Time Outdoors`) showed negligible absolute and relative bias (ranging from -1.4% to -3.7%) and near zero variability (Table 3.4), confirming the stability and accuracy of fixed-effect estimation under the pooled analysis.

For the random effects, the Subject-level variance (`u_subject`) was estimated at 1.503 (bias = -0.497; relative bias = -24.9% MSE = 0.247), but with the variance across repetitions near zero (variance = 0.000), this reflected a consistent underestimation. The resulting MSE (0.247) was driven almost entirely by bias rather than sampling variability.

The Site-level variance (`u_site`) was estimated at 2.980 (SD = 1.554; bias = -0.020; relative bias = -0.7%; MSE = 2.415), showing a bias that is close to zero but with high variability across repetitions. The high variance indicates that the estimates were very unstable across the simulations, even though the mean estimates are close to the true value. The high MSE reflects this sampling variability rather than systematic bias.

The Residual variance was recovered at 196.64 (SD = 0.0004; bias = 0.641; relative bias = 0.3%; MSE = 0.411). Both bias and relative bias were minimal, indicating accurate recovery of the residual variance. Variability across repetitions was close to zero, suggesting that residual variance estimates were highly stable across the simulations.

In summary, the fixed effects were accurately recovered with negligible bias, low relative bias, and small mean squared errors, indicating stable and reliable estimates. Among the random effects, the subject-level variance showed consistent underestimation , while the site-level variance was recovered with minimal bias but considerable between-repetition variability. The residual variance was closely estimated to the true value with very small bias. Overall, the simulation results showed good recovery performance with most discrepancies arising from underestimation of the subject-level variance and variability in the site-level estimates. These pooled results provide a quantitative baseline for evaluating the parameter recovery performance of FSDA.

The full R code and package information used to perform the 1,000 repetition pooled analysis is provided

in Appendix A.4 to ensure reproducibility.

3.5.2 FSDA: Synthetic Data

The synthetic datasets generated under FSDA were analyzed across 1,000 repetitions. As in the gold standard pooled analysis, descriptive statistics for all fixed and random effects are presented in Table 3.6.

Table 3.6: *Descriptive statistics for fixed and random effects under FSDA pooled synthetic data (1,000 simulation repetitions). Mean, SD, and variance are reported.*

Parameter	Mean	SD	Variance
Fixed Effects			
Intercept	46.097	0.185	0.034
Sex	1.530	0.124	0.016
Age	-0.484	0.017	0.000
Time Outdoors	1.468	0.033	0.001
Random Effects			
Subject-level variance	1.984	0.440	0.194
Site-level variance	2.374	0.188	0.035
Residual			
Residual variance	196.162	1.121	1.256

Performance metrics (bias, relative bias, variance, MSE) relative to the true generating values are reported in Table 3.7. These results provide a quantitative assessment of estimation accuracy for the FSDA-generated datasets, serving as a basis for comparison with the gold standard pooled analysis presented in Section 3.5.1.

Table 3.7: *True values, mean estimates, bias, relative bias, variance, and MSE for fixed and random effects under FSDA pooled synthetic data.*

Parameter	True Value	Mean Estimate	Bias	Rel. Bias (%)	Variance	MSE
Fixed Effects						
Intercept	45.0	46.097	1.097	2.4	0.034	1.236
Sex	2.0	1.528	-0.472	-23.6	0.016	0.238
Age	-0.5	-0.484	0.016	-3.2	0.000	0.001
Time Outdoors	1.5	1.468	-0.032	-2.1	0.001	0.002
Random Effects						
Subject-level variance	2.0	1.984	-0.017	-0.8	0.194	0.194
Site-level variance	3.0	2.374	-0.626	-20.9	0.035	0.427
Residual						
Residual variance	196.0	196.162	0.162	0.1	1.256	1.282

For the fixed effects, parameter estimates were generally well recovered with small absolute and relative bias, also noted were low variability across repetitions. The `Intercept` coefficient was slightly overestimated (mean = 46.10; bias = 1.095; relative bias = 2.4%; MSE = 1.236) with low variance (0.034) indicating stable recovery across simulations. The `Sex` coefficient showed the highest deviation (mean = 1.528; bias = -0.472; relative bias = -23.6%; MSE = 0.238), this shows a noticeable but stable underestimation, as its

variance (0.016) is near to zero. The continuous covariates, `Age` and `Time Outdoors`, were accurately recovered, with minimal bias (0.016 and -0.032) and small relative bias (-3.2% and -2.1%). Both had very low variance and MSE values, suggesting highly stable estimates across repetitions.

For the random effects, the Subject-level variance (`u_subject`) was estimated at 1.984 (bias = -0.017;relative bias = -0.8%; variance = 0.194; MSE = 0.194), showing very close agreement to the true value (2.0). The variance (0.194) and MSE values (0.194) being close to zero indicates accurate and stable estimation across repetitions. The Site-level variance (`u_site`) was estimated at 2.374 (bias = -0.626; relative bias = -20.9%; variance = 0.035; MSE = 0.427), indicating noticeable underestimation. Although the variability across repetition was low, the mean estimate remained systematically biased, indicating consistent underestimation. The Residual variance was recovered at 196.162 (bias = 0.162; relative bias = 0.1%; variance = 1.256; MSE = 1.282). Estimates were centered closely around the true value (196.0), with very small relative bias. The slightly larger variance compared to other parameters indicates mild dispersion between repetitions, but overall recovery remained highly accurate.

Overall, the FSDA results showed accurate recovery of most fixed-effects, with small biases and low MSE values across repetitions. However, a noticeable underestimation was observed for the `Sex` coefficient and the site-level variance. In the next subsection, these results are compared to the results from the gold standard pooled analysis.

The R code used to generate the synthetic datasets and conduct the FSDA simulation study is provided in Appendix A.5.

3.5.2.1 Direct Comparison FSDA vs Gold Standard

To directly compare FSDA with the gold standard pooled analysis, Table 3.8 presents the mean estimates, bias, relative bias, variance, and MSE for all parameters, while Figures 3.6, 3.7 and 3.8 provide corresponding graphical summaries. Together, these results offer a comprehensive comparison of model performance across all parameters. Numerical summaries for individual parameters are detailed in Sections 3.5.2 and 3.5.1. The lollipop plots in Figures 3.6, 3.7 and 3.8 display the bias, variance,and MSE for all parameters, facilitating a clear visual comparison between FSDA and the gold standard pooled analysis.

Table 3.8: *Side-by-side comparison of mean estimates, bias, relative bias, variance, and MSE between the gold standard pooled analysis and FSDA across all parameters.*

(a) *Mean estimate and Bias (part a)*

Parameter	True Value	Mean (Pooled)	Mean (FSDA)	Bias (Pooled)	Bias (FSDA)
Fixed Effects					
Intercept	45.0	45.084	46.097	0.084	1.097
Sex	2.0	1.927	1.528	-0.073	-0.472
Age	-0.5	-0.488	-0.484	0.012	0.016
Time Outdoors	1.5	1.479	1.468	-0.021	-0.032
Random Effects					
Subject-level variance	2.0	1.503	1.984	-0.497	-0.017
Site-level variance	3.0	2.980	2.374	-0.020	-0.630
Residual					
Residual variance	196.0	196.641	196.162	0.641	0.162

(b) *Relative Bias, Variance and MSE (part b)*

Parameter	Rel. Bias (%) (Pooled)	Rel. Bias (%) (FSDA)	Var (Pooled)	Var (FSDA)	MSE (Pooled)	MSE (FSDA)
Fixed Effects						
Intercept	0.19	2.4	0.374	0.034	0.381	1.236
Sex	-3.66	-23.6	0.000	0.002	0.005	0.238
Age	-2.39	-3.2	0.000	0.000	0.000	0.001
Time Outdoors	-1.40	-2.1	0.000	0.001	0.000	0.002
Random Effects						
Subject-level variance	-24.86	-0.8	0.000	0.194	0.247	0.194
Site-level variance	-0.68	-20.9	2.415	0.035	2.415	0.427
Residual						
Residual variance	0.33	0.1	0.000	1.256	0.411	1.282

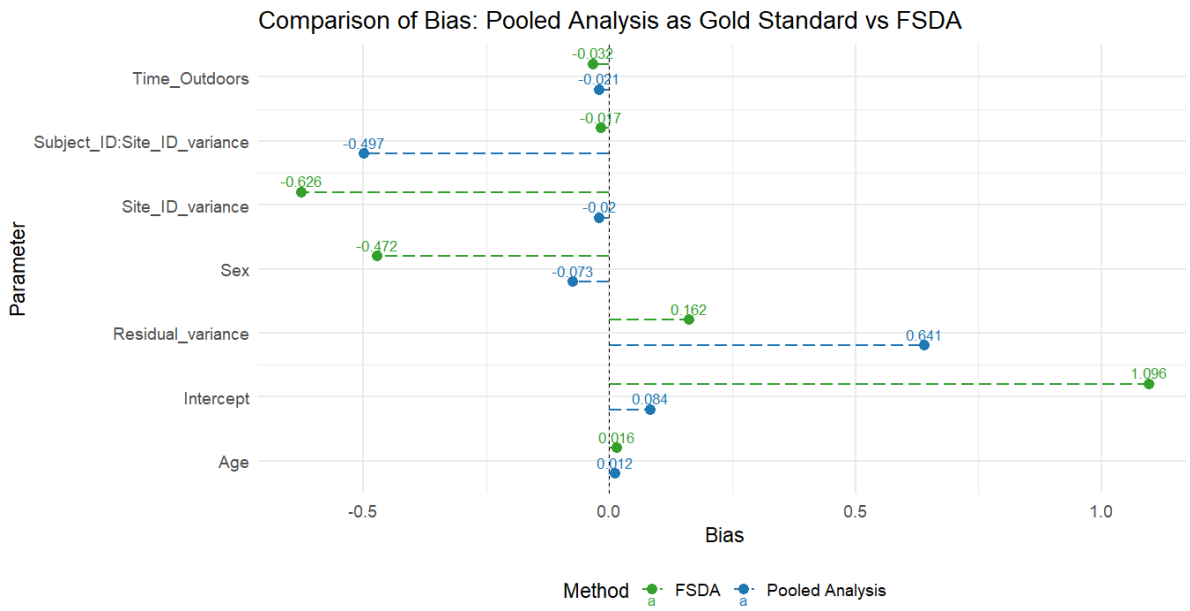


Figure 3.6: *Bias of parameter estimates across 1,000 repetitions for pooled analysis and FSDA. Each lollipop represents a parameter estimate, with pooled analysis (blue) and FSDA (green) shown side by side. Bias close to zero indicates accurate recovery of the true values.*

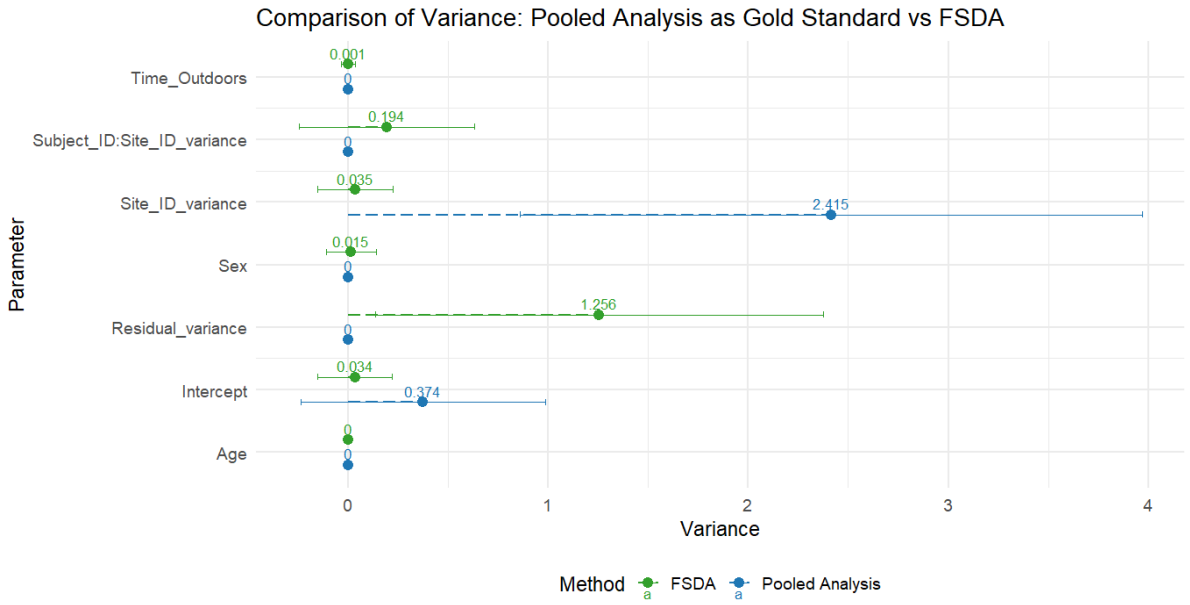


Figure 3.7: *Variance of the parameter estimates across 1,000 repetitions for pooled analysis and FSDA. Lollipops represent the variance of each estiamte, with wiskers showing ± 1 standard deviation. Pooled analysis (blue) and FSDA (green) are displayed side by side to facilitate comparison of variability.*

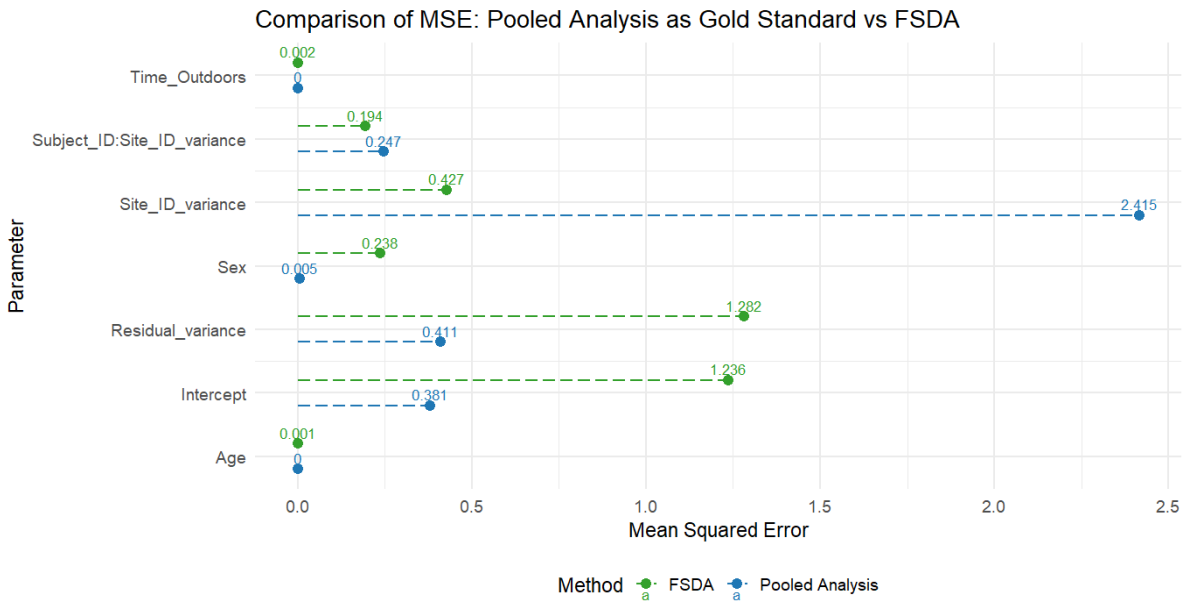


Figure 3.8: *MSE of parameter estimates across 1,000 repetitions for pooled and FSDA. Each parameter is represented by paired lollipops, with pooled analysis (blue) and FSDA (green). Lower MSE values indicate better overall performance, balancing bias and variance.*

Across the simulation repetitions, both the pooled analysis and the FSDA approach recovered the fixed effect coefficients for `Age` and `Time Outdoors` with negligible bias, very small variance and low MSE values, indicating accurate and stable recovery of these continuous predictors. These predictors showed the most stable performance across both methods.

For the `Sex` coefficient, both methods produced underestimates, but the bias was more pronounced in FSDA (bias = -0.472; relative bias = -23.6%) compared with the gold standard pooled analysis (bias = -0.073; relative bias = -3.66%). The `Intercept` also differed slightly between methods; both approaches slightly overestimated the true value of 45.0, but the FSDA estimate (46.097) showed greater bias and higher MSE than the pooled analysis.

For the random effects, the subject-level variance was underestimated in the gold standard pooled analysis (mean = 1.503; bias = -0.497; relative bias = -24.86%), while FSDA recovered this parameter with minimal bias (mean = 1.984; bias = -0.017; relative bias = -0.8%). In contrast, the site-level variance showed small bias but large variability in the pooled analysis (mean = 2.980; variance = 2.415; relative bias = -0.68%), whereas FSDA produced a more stable but noticeably underestimated estimate (mean = 2.374; bias = -0.626; relative bias = -20.9%).

The residual variance was recovered accurately by both methods. The pooled analysis showed minimal bias (relative bias = 0.33%), while FSDA showed minimal bias (relative bias = 0.1%) but with slightly higher variability across repetitions.

Overall, the two methods showed similar estimates for most fixed effects and for the residual variance, but differed in the estimation of variance components. The gold standard pooled analysis showed instability

(high variability across repetitions) in the site-level variance estimation, while FSDA showed a consistently biased estimate for the site-level variance and the `Sex` coefficient. They also differed in the recovery of the subject-level variance where the pooled analysis showed consistent underestimation, while FSDA recovered this parameter with minimal bias.

Having evaluated the statistical performance of FSDA relative to the gold standard pooled analysis, we next assessed the disclosure risk associated with the synthetic datasets. While synthetic data aim to preserve the utility of the original data, it is also critical to ensure that individual-level information cannot be re-identified. In the following subsection, we present the results of the disclosure risk assessments for the FSDA-generated datasets.

3.6 Disclosure Risk Assessment

Before assessing disclosure risk, the continuous variables (`Age`, `Time Outdoors`, and `Vitamin D level`) were grouped into bins to ensure consistent comparison of quasi-identifiers and target variables across records. This binning was performed automatically by the `multi.disclosure()` function in the `synthpop` package, which creates categorical bands based on the distribution of each variable. This approach provides a realistic basis for assessing disclosure risk without distorting the underlying data distributions.

Disclosure risk was evaluated separately for each of the eight sites using the quasi-identifiers `Site_ID`, `Sex`, `Visit`, `Age`, `Time Outdoors`, along with three sensitive target variables: `Subject_ID`, `Vitamin D level`, and `Deficient`. The measures computed included (UiO) and (RepU) for identity disclosure, and Dorig and DiSCO for attribute disclosure (see Section 2.6.2.7).

Table 3.9: *Disclosure risk assessment by site: Identity and attribute disclosure (%).*

Measure	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
Identity Disclosure								
UiO	0.03	0.02	0.21	0.01	0.04	0.04	0.03	0.02
RepU	0.01	0.01	0.00	0.00	0.01	0.01	0.01	0.00
Attribute Disclosure: Subject_ID								
Dorig	0.03	0.02	0.21	0.01	0.04	0.04	0.03	0.02
DiSCO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Attribute Disclosure: Vitamin D Level								
Dorig	0.03	0.03	0.31	0.02	0.04	0.05	0.03	0.04
DiSCO	0.00	0.00	0.03	0.05	0.00	0.00	0.01	0.01
Attribute Disclosure: Deficient								
Dorig	0.03	0.03	0.80	0.03	0.05	0.05	0.09	0.28
DiSCO	0.13	0.03	0.18	0.20	0.00	0.00	0.04	0.00

Note. UiO = Unique in Original; RepU = Replicated Uniques; Dorig = Disclosive in Original; DiSCO = Disclosive in Synthetic Correct Original. All values represent percentages (%).

Across all sites, both identity and attribute disclosure risks were minimal (Table 3.9). For identity disclosure, the proportion of unique records in the original data (UiO) ranged from 0.01% to 0.21%, while replicted unique in the synthetic data (RepU) were almost entirely absent, with values between 0% and 0.01%. For

attribute disclosure, the target variable `Subject_ID` exhibited consistently low risk across all sites, with Dorig values up to 0.21% and no disclosive matches (DiSCO = 0%) in the synthetic data. Slightly higher, yet still negligible, risks were observed for `Vitamin_D_level`, where Dorig ranged from 0.02% to 0.31% and DiSCO values from 0% to 0.05%. For `Deficient`, attribute disclosure risk was somewhat higher in the original data (Dorig = 0.03% to 0.80%) but remained low in the synthetic data (DiSCO = 0% to 0.20%).

Overall, these results demonstrate that both identity and attribute disclosure risks remained extremely low across all sites and target variables, confirming the effectiveness of the FSDA approach in mitigating re-identification and sensitive attribute inference risks (see Table 3.9). The full R code used for this assessment is provided in Appendix A.6.

Taken together, the results presented across the preceding subsections collectively demonstrate that the FSDA framework effectively balances analytical validity and confidential protection. The synthetic datasets achieved high statistical utility, produced parameter estimates that are not far off aligned with those from the original pooled data, and maintained negligible risks of identity and attribute disclosure. The balance highlights the potential of FSDA as a practical approach for generating analytically robust yet privacy-preserving datasets in multi-site longitudinal studies.

In the next chapter, these results are interpreted in the broader methodological context. We discuss potential sources of observed variation, evaluate the implications of the findings for federated synthetic data in multilevel analyses, and reflect on the broader utility-privacy trade-off underpinning this framework.

4 Discussion

4.1 Summary of Key Findings

In this thesis, we set out to evaluate whether FSDA applied to LMM in a multi-site context, can reproduce inferential results comparable to those obtained from the gold standard pooled analysis, while maintaining data privacy. Overall, the results indicate that FSDA provides good analytical utility and supports valid inference for multilevel data, while maintaining a strong level of data privacy protection, and although some differences emerged in the recovery of variance components.

The simulated datasets (original data) successfully reproduced realistic features of a multi-site hierarchical data, including heterogeneity between sites, within-subject clustering, and realistic covariate-outcome relationships. This provided solid foundation for evaluating the performance and robustness of FSDA in a hierarchical modeling context.

The synthetic datasets generated under FSDA broadly reproduced the marginal and joint distributions of the original simulated data. Stratification by site and nested subject identifiers ensured that the multi-level structure was preserved, while variable-specific synthesis methods produced distributions that aligned with the underlying data-generation mechanisms.

We also see that the synthetic data demonstrated strong utility performance, with similarity in distribution to the original data. Both variable-specific and overall utility measures indicated that our synthesizing model preserved essential statistical properties of the original datasets, with pMSE and S_pMSE values close to their expected null benchmarks. This finding suggest that synthetic data created can preserve analytical validity.

Across the simulation of 1000 repetitions, both FSDA and the gold standard pooled analysis accurately recovered the fixed effect coefficients, particularyy `Age` and `Time Outdoors`, which consistently showed negligible bias and minimal variability. The `Sex` effect was recovered with low to moderate bias in both methods, though FSDA showed a more pronounced underestimation. The `Intercept` exhibited positive bias in both methods, with FSDA showing slightly larger deviation from the true value.

For the random effects, the subject-level variance was recovered accurately and precisely under FSDA with negligible bias, whereas the gold standard pooled analysis consistently underestimated this component across repetitions. For the site-level variance, the gold standard pooled analysis produced estimates with negligible bias, but displayed substantial variability. In contrast, FSDA produced consistently biased estimates.

Finally, the privacy risk assessment, confirmed that the synthetic datasets maintained low levels of membership disclosure risk, well within acceptable thresholds. This demonstrates that FSDA can effectively safeguard individual-level information while still allowing valid statistical inference.

Overall, the findings support FSDA as a promising framework for privacy-preserving multi-site analysis. Across the simulations, FSDA maintained good analytical fidelity to the gold standard pooled analysis, preserved the key multilevel structure of the original data, and generated synthetic data with high utility and low disclosure risk. Based on the conditions induced by the low signal to noise (SNR), differences emerged in

the estimation of certain variance components. FSDA demonstrated robust recovery of core model components, including subject-level variance, while differences remained in the estimation of higher level variance components without undermining the overall inferential validity of the synthetic data.

Having established these main results, the next section interprets these findings within the context of existing literature, discusses why certain discrepancies emerged between FSDA and the gold standard pooled analyses, and explores how the present results align with or differ from previous research on synthetic data generation and federated analysis.

4.2 Interpretation and Comparison to Existing Literature

The results of this thesis highlight several important methodological insights regarding the behaviour of FSDA relative to the gold standard pooled analysis when applied to complex multilevel or hierarchical data. The observed similarities and differences can be understood within the theoretical frameworks of mixed effects models, synthetic data generation and tree-based methods.

4.2.1 Fixed Effects and Core Model Structure

The close agreement between FSDA and the gold standard pooled analysis for most fixed effects coefficients is consistent with earlier work that showed that synthetic data generation methods can reliably reproduce main associations when relationships are strong and well supported by the data (Drechsler 2011; Reiter 2003). The use of CART (synthesizing model) in FSDA, enables it to capture complex nonlinear patterns, interactions and conditional dependencies, helping to preserve the structure and key inferential properties of the original data. This aligns with prior literature, were decision-tree methods effectively propagate conditional relationships across distributed data sources (Azizi et al. 2023; Nowok et al. 2016; Snoke et al. 2018).

The high relative bias observed for `Sex` in the FSDA method can be attributed to choice of synthesizing model rather than weakness of FSDA. In the synthesis process, we generated `Sex` using a logistic regression model rather than a CART synthesizing model. Prior work by Hayes et al. (2015), noted that logistic models can produce higher bias than tree-based methods in nonlinear or interaction-rich settings, suggesting that the observed bias in `Sex` likely reflects limitations of the logistic model used for synthesis, rather than a failure of FSDA to reserve fixed effects relationships.

4.2.2 Subject-Level Variance and the Limitations of Low Signal to Noise Ratio

It was observed that the gold standard pooled analysis consistently underestimated the subject-level variance. This behaviour is consistent with known limitations of variance component estimation in LMM under low SNR conditions, were substantial residual variability dominates the total outcome variance. In such settings, separating true subject-level variance from residual variance becomes statistically challenging, even when

the model is correctly specified. The difficulty of variance decomposition under low SNR has been widely documented in the mixed effect modelling literature, particularly when residual variance is large relative to the random effects variance (Pinheiro and Bates 2004b; Snijders and Bosker 2012; Verbeke and Molenberghs 2011).

In contrast, FSDA showed improved robustness in recovering subject-level variance in the low SNR setting. This behaviour can be attributed to the properties of the CART synthesizing model, which approximates the conditional distribution of the outcome by recursively partitioning the predictor space. In the presence of substantial residual variability and complex outcome structure, CART is less prone to over-partitioning and produces broader terminal leaves that preserves realistic within-leaf variability. The subsequent Bayesian bootstrap within leaves therefore generates synthetic data that retain subject-level heterogeneity more effectively under noisy conditions. This robustness of tree-based synthesis methods in complex, high noise settings has been noted in the synthetic data literature (Drechsler and Reiter 2011; Nowok et al. 2016; Reiter 2005c; Snoke et al. 2018).

As a sensitivity analysis, alternative residual variance specification corresponding to higher signal to noise ratios were explored in preliminary analyses. These results indicated that, under higher signal to noise conditions, the gold standard pooled analysis recovered subject-level variance with minimal bias, consistent with theorectical expectations, whereas FSDA using CART based synthesizing model showed evidence of overfitting due to overly fine partitions of the predictor space. Given the focus of this thesis on evaluating FSDA robustness in complex, realistic and challenging data conditions, the primary analyses were conducted under the low SNR and the high signal to noise scenario were not pursued further and are discussed as a limitation.

4.2.3 Site-Level Variance: Few Clusters and Method Specific Behaviour

The gold standard pooled analysis was able to reproduce a mean estimate that was close to the true value, but the estimates were very unstable (varied widely) across simulation repetitions. This consistent downward shift is consistent with the small-cluster bias described by McNeish and Stapleton (2016) , who noted that mixed-effects models tend to underestimate higher-level variance components when the number of cluster is limited. Given that both analyses included only eight sites, such bias is expected even under correctly specified models. The slightly larger underestimation in the FSDA analysis likely reflects the mild smoothing behaviour of tree-based synthesizers, which predict constant values within terminal nodes and may compress between-cluster variability if the trees do not fully partition along site boundaries or if pruning limits depth (Breiman et al. 2017; Drechsler and Reiter 2011; Hastie et al. 2009; Reiter 2005c).

4.2.4 Residual Variance and Overall Error Structure

The residual variance was recovered reasonably well by both methods, with just small differences relative to the true value. The slight higher variability seen in FSDA's residual variance is consistent with the idea that smoothing of global structure may shift a small amount of unexplained variability into the residual component

(Breiman et al. 2017; Hastie et al. 2009). Importantly, both bias and MSE were small in absolute and relative terms, suggesting limited practical consequence for inference.

4.2.5 Utility of the Synthetic Data

The utility result showed that the generated synthetic data was structurally similar to the original data. This aligns with previous findings from earlier work by Snoke et al. (2018) and Nowok et al. (2016), who showed that when a synthetic method is able to preserve the structure and relationships in the data, the classifier used in the pMSE framework struggles to distinguish synthetic records from original records. This usually produces low pMSE values, which is exactly what we observed for most of the variables.

The standardized overall S_pMSE could not be computed, as the empirical resampling procedure required for estimating the null distribution was disabled (`resampling = "none"`) due to computational constraints associated with the high dimensionality of the dataset and the number of replications required. This approach is not unusual, and is consistent with recommendations by Snoke et al. (2018), who noted that direct computation of raw pMSE alone still provides a valid basis for interpretation, as long as the synthesis approach remains consisten across datasets.The utility results of this thesis do follow this pattern, and even without S_pMSE values for the overall utility, the pMSE was small and indicated good overall utility.

For variable-specific utility, the large S_pMSE observed for the site-level random effect (`u_site`) also aligns well with previous findings. This discrepancy is expected when a variable has few categories or hierarchical structures, as is the case in our thesis with only eight sites (Nowok et al. 2016; Raab et al. 2021; Snoke et al. 2018). Consequently, large S_pMSE values for such variables do not necessarily indicate poor synthesis quality but rather reflect the limited discriminative power of the propensity model for low-dimensional categorical predictors (Nowok et al. 2016; Raab et al. 2021; Snoke et al. 2018). The visual comparisons showed strong agreement between original and synthetic distributions, supporting the view that the inflated S_pMSE reflects structural characteristics rather than true synthesis error.

Overall, the utility results in this thesis were consistent with previous literature; synthesis model were able to preserve both marginal and joint distributions effectively, and the deviations seen in the site-level random effect (`u_site`) can be attributed to structural characteristics of the data rather than reflecting synthesis error. This findings support the conclusion that the generated synthetic data successfully maintained high level of statistical utility and suitable for downstream analysis.

4.2.6 Disclosure Risk

The disclosure risk results showed that the synthetic datasets generated under FSDA carried very low risks of both identity disclosure and attribute disclosure across all the sites. This is fully consistent with what has been reported in synthetic data literature for synthetic methods that rely on tree-based methods such as CART.

First, the very low values for identity disclosure risk (UiO and RepU) match earlier findings by Nowok et al. (2016), Reiter (2005a), and Raab et al. (2024), who showed that synthetic datasets usually contain few, if any, records that match original datasets closely enough to be regarded as "unique". Because synthetic datasets does not copy or reproduce the exact records of the original datasets, exact or near duplicates are rare. This charateristic explains why RepU were virtually zero across all eight sites.

These results also support the general understanding that identity disclosure is limited in fully synthetic data; since no original record appears in the synthetic dataset, an intruder cannot be sure if any specific individual was part of the original data. The negligible RepU values observed in this thesis do confirm with the theory.

The low attribute disclosure risks we observed also aligns well with previous findings which has shown that tree-based methods or synthesizers such as CART, do not copy or reproduce exact individual records; instead it learns or captures broad conditional relationships between variables (Drechsler 2011; Nowok et al. 2016; Raghunathan et al. 2003). This behaviour helps to explain why the target variables all showed very low DiSCO values, even when the synthetic records appear similar to the original records.

This "smoothing" effect by CART, is particularly beneficial in the concept of attribute disclosure as CART is able to preserve the overall structure of the data while avoiding disclosing exact details of the original data. This reduces the chance that the synthetic data could reveal sensitive information about an actual individual, making CART an effective tool for limiting attribute disclosure.

In summary, the disclosure results closely align with expectations from the synthetic data and privacy literature. They confirm that FSDA, when implemented with CART synthesizer, provides a strong level of confidentiality protection. Together with the strong utility findings, these results support the view that FSDA is appropriate for privacy-preserving multi-site analyses.

4.3 Methodological Considerations

Several characteristics of the data-generating process help explain the patterns observed in the results, particularly with respect to the estimation of variance components.

First, the simulation was intentionally designed to operate under a low signal to noise ratio, induced by a relatively large residual variance. In such setting, linear mixed effects models face a well known issue of separating subject-level variance from residual variance, even when the model is correctly specified. When residual variance dominates the total outcome variance, the ability to identify random-effects variances is substantially reduced, leading to biased or unstable variance component estimates.This phenomenon is well documented in the mixed effects modelling literature (Pinheiro and Bates 2004b; Snijders and Bosker 2012; Verbeke and Molenberghs 2011).

Second, the dataset contained only eight sites, therefore placing the site-level random effect in a recognized "small-cluster" setting. It has been shown by previous work that random-effects variance estimation with a small number of higher-level clusters can be unstable and highly sensitive to samplying variability, even

when cluster sizes are large (McNeish and Stapleton 2016). This explains the substantial variability observed in site-level variance estimates from the gold standard pooled analysis.

With respect to FSDA, the observed behaviour of the CART synthesizing model reflects the inherent nature of tree based models. CART approximates the conditional distribution of the outcome by prioritising splits that yield the greatest reduction in within-node variability, without explicitly modelling hierarchical or variance components. In settings with limited number of clusters, the site indicator often provides relatively weak predictive improvement. As a result, CART may not fully partition the data along site boundaries, leading to terminal leaves that pool observations from multiple sites. The subsequent Bayesian bootstrap within such leaves smooths site-specific differences, resulting in attenuated between-site variability in the synthetic data. This behaviour is consistent with known limitations of tree-based synthesis methods in recovering higher level variance components when the cluster counts are small.

Taken together, these considerations demonstrate that both modelling and synthesis outcomes are strongly influenced by the structure of the data, with particular reference to the low signal to noise ratio and the small number of higher-level clusters (number of sites). These results highlight the importance of aligning both modelling and synthesis strategies with the structural properties of the data when evaluating FSDA inferential and privacy preserving frameworks.

4.4 Limitations

Several limitations should be considered when interpreting the results of this thesis.

4.4.1 Low Signal to Noise Ratio

The simulation was intentionally designed to operate under a low signal to noise ratio, induced by a relatively large residual variance to reflect real-world evidence settings. While this choice enabled a meaningful assessment of FSDA robustness under complex and noisy conditions, it also limits the generalisability of the results to settings with stronger individual-level signal. In particular, variance component estimation in both pooled mixed effects models and FSDA is known to be challenging when residual variability dominates the total variance, and performance may differ under higher signal to noise setting.

4.4.2 Small Number of Clusters

The dataset included only eight sites, placing the site-level random effect in a small cluster setting. Variance component estimation from very small clusters are inherently unstable. This limitation affected both methods, with the gold standard pooled analysis showing high variability, while in FSDA, the CART synthesizing model tended to smooth site-specific differences, leading to underestimation of between-site variance. These results may not generalise to settings with a higher number of sites.

4.4.3 Synthesizer Model Choices in FSDA

Different synthesizer models were used for different variables, reflecting a variable-specific synthesis strategy. Notably, the `Sex` variable was synthesized using logistic regression rather than CART, which introduced bias in the estimate. This highlights that FSDA performance can be sensitive to the choice of the synthesizer method chosen and underscores the importance of careful model selection and sensitivity analyses when implementing synthetic data generation.

4.4.4 Simplified Data-Generating Mechanism

The simulation of the data was solely based on a basic random-intercept structure without random slopes, interactions, or more complex hierarchical dependencies. FSDA performance may behave differently in scenarios involving richer longitudinal or higher hierarchical structure often found in real-world data, which warrants further investigation.

Overall, these limitations indicate that the results of this thesis should be interpreted within the specific simulation conditions and not assumed to generalize to all multilevel or federated analysis settings.

4.5 Practical Implications

Despite these constraints, several practical insights did emerge from this thesis.

- FSDA provides a promising viable alternative to gold standard pooled analysis in settings where centralized data pooling is not feasible due to privacy, legal, or regulatory constraints.
- Under the simulation study conditions considered, FSDA preserved the essential multilevel structure of the original data, enabling reliable recovery of fixed effects.
- Estimation of variance components must be interpreted carefully, as both gold standard pooled analysis and FSDA showed limitations arising from the small number of clusters (sites) and low signal to noise ratio, highlighting that variance component recovery remains challenging in such settings regardless of the analytical framework.
- The choice of synthesizing model plays a critical role in FSDA performance. The observed bias in the `Sex` coefficient illustrates that an inappropriate synthesizer can adversely affect inferential accuracy.
- For organisations that require privacy-preserving multi-site analysis, FSDA offers a practical balance between confidentiality protection and analytical validity, particularly when inference focuses on fixed effects and aggregate conclusions rather than precise recovery of higher level variance components.

Overall, these results suggest that FSDA is a promising alternative well suited to collaborative health research scenarios where sharing of original individual-level data is restricted, and provided that its limitations, especially regarding variance components, are acknowledged.

4.6 Future Work and Recommendations

This Thesis identifies several directions for further work.

4.6.1 Expand the Number of Clusters

Exploring data with 15 or more sites would enable a more stable evaluation of site-level variance recovery across both gold standard pooled analysis and FSDA method.

4.6.2 Investigate Alternative Synthesizing Models

Future work could systematically compare FSDA performance across a broader range of synthesis models to better understand how different modelling assumptions affect the recovery of longitudinal or hierarchical structures and variance components. Candidate approaches could include:

- **Tree-based methods**, such as fully CART-based synthesis, pruned CART, and RF synthesizers, which may offer improved handling of nonlinear relationships and interactions.
- **Probabilistic and parametric models**, including other parametric synthesizers that explicitly model joint distributions, and Bayesian Networks (BNs), which represent conditional dependencies through graphical structures may offer advantages of preserving hierarchical relationships and variance components in multi-site data.
- **Deep learning approaches**, such as generative adversarial networks (GANs), variational autoencoders (VAEs), and diffusion-based models, which have shown promise in capturing complex multivariate relationships in high-dimensional data.
- **Hybrid or ensemble approaches**, combining strengths of different models (e.g., CART with RF, or GANs with tree-based components (e.g., adversarial random forest)), which may offer some level of flexibility.

Comparative evaluation of these approaches could help identify models that better capture hierarchical structure and reduce bias in variance components, and improve robustness under challenging settings such as small numbers of clusters or low signal to noise ratios.

4.6.3 Evaluate FSDA in Additional Modelling Contexts

Future investigation in FSDA should assess other modelling frameworks beyond linear mixed effects models considered in this thesis. This includes more complex mixed effects models, non-linear models, Bayesian hierarchical models, and models for generalized or survival outcomes. This would help determine if FSDA performance observed in this thesis generalizes beyond the specific modelling context considered. Such extensions would provide a more clearer understanding of the robustness and scope of FSDA across diverse analytical contexts.

4.6.4 Apply FSDA to Real Clinical or Public Health Data

Applying FSDA to a real clinical or public health data would help validate practical feasibility, utility, fidelity and privacy performance beyond just simulation settings, complementing simulation based results.

4.6.5 Alternative Estimation Frameworks for Variance Components

An important direction for future work will be to investigate the estimation of variance components in hierarchical models with small number of clusters. In this thesis, we estimated variance components using REML, which is a well known method but seems to exhibit instability and poor estimation when the number of higher level clusters is small (McNeish and Stapleton 2016).

Bayesian hierarchical models (BHM) may provide an alternative by incorporating weak informative priors on variance components, which could allow for partial pooling and more stable inference in small cluster settings. Applying Bayesian variance estimation within both pooled and FSDA frameworks may improve the recovery and uncertainty quantification of higher level variance components (Studnicki and Piepho 2024). Future work could look into such approaches in reducing the instability and bias observed under REML estimation in multi-site synthetic data analyses.

5 Conclusions

In this thesis, we evaluated the performance of FSDA in a multi-site linear mixed effects modelling context, with the goal of determining whether synthetic data generated and pooled in a federated setting can reproduce the inferential results obtained from a gold standard pooled analyses while also simultaneously protecting confidential individual-level data privacy. Using simulated hierarchical (multi-level) data with eight sites and two repeated measurements per subject, FSDA was systematically compared with the gold standard pooled analysis in terms of statistical utility, parameter recovery, and privacy risk.

Overall, the results demonstrate that FSDA is capable of achieving good analytical fidelity under the conditions considered. The synthetic datasets preserved key relationships among covariates and retained the multi-level structure of the original data. Fixed effects were recovered with negligible bias, supporting the use of FSDA for primary inferential objectives in multi-site health research. For subject-level variance, FSDA showed reduced bias relative to the gold standard pooled analysis under low signal to noise conditions, likely reflecting the ability of CART synthesizing model to preserve within-subject dependence through conditional modelling, even when the residual variability is high.

Differences between FSDA and the gold standard pooled analysis were most noticed in the estimation of the site-level variance. FSDA produced an underestimated site-level variance estimate, while the gold standard pooled analysis showed substancial unstable variance across simulations. These behaviours can be attributed to the limited information available for estimating higher-level variance components with a small number of clusters, rather than deficiencies of either method.

The utility assessment confirmed that the synthetic data remained very similar to the original data across both marginal and joint distributions. With the variable-specific pMSE and S_pMSE values we were able to come to the conclusion that the synthetic data was able to retain the statistical properties of the original dataset, with minor deviations attributable to structural characteristics of the data rather than synthesis deficiencies. Importantly, the disclosure risk analysis showed very low levels of identity and attribute disclosure across all the sites and target variables. This indicates that FSDA can reduce re-identification risk while still able to produce data suitable for meaningful statistical analysis.

In summary, the results indicate that FSDA offers a promising practical and effective approach for privacy-preserving multi-site analysis. It allows valid statistical inference to be made without requiring the original data to be centrally pooled, making it well suited for health research settings that can be constrained by confidentiality, ethics, or regulatory requirements. Although limitations such as small cluster numbers, low signal to noise ratio, and method-specific synthesis choices should be considered when interpreting the results, the overall performance observed in this thesis highlights the promising potential of FSDA for real world applications.

Future work should extend this evaluation to scenarios with richer longitudinal structures, larger number of clusters, different synthesizing models, and broader analytical objectives. Applying FSDA to real world multi-site datasets will further provide a deeper insight into its practical strengths and limitations. Neverthe-

less, this thesis provides a clear evidence that FSDA is a promising framework for enabling collaborative, privacy-conscious statistical analysis in multi-site health research environments.

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Appendices

A Appendix

This appendix provides supplementary material to support the main thesis. Specifically, it includes:

- The R code used for simulating the multi-site longitudinal dataset (Appendix A.1);
- The R code used for generating synthetic datasets using the synthpop package (Appendix A.2);
- Additional figures and diagnostics that complement the main text but were not included in the Results section for brevity.

These materials ensure transparency and reproducibility of the analyses described in the thesis.

A.1 Appendix A. Simulated Data Code

This appendix provides the R code used to genarate the simulated multi-site longitudinal dataset described in Section 2.6.2 and summarized in Section 3.2.

The code defines site-level and subject-level structures using the `simstudy` package, generates covariates (`Age`, `Sex`, `Time Outdoors`), and computes the outcome variables (`Vitamin D level`) according to the specified linear mixed-effects model.

```
# ------------------------
# 1. Load necessary packages
# ------------------------
library(simstudy)       # for data simulation
library(lme4)           # for fitting mixed models
library(tidyverse)      # for data manipulation and visualization
library(data.table)     # data manipulation and handling of large datasets

# ------------------------------
# 2. Defines site-level structure
# ------------------------------
gen.site <- defData(varname = "u_site",
                    dist = "normal",
                    formula = 0,
                    variance = 3,
                    id = "Site_ID")
```

```r
# Add the number of subjects per site
gen.site <- defData(gen.site,
                    varname = "nSubjects",
                    formula = 80000,
                    variance = .3,
                    dist = "clusterSize")

# Generate the site-level data
set.seed(282721)

dtSite <- genData(8, gen.site, id = "Site_ID")

# View the site-level data
dtSite

# ----------------------------------
# 3. Defines subject-level structure
# ----------------------------------
# Define subject-level data
gen.subject <- defDataAdd(varname = "u_subject",
                          dist = "normal",
                          formula = 0,
                          variance = 2)
# Add the subject-level variables
gen.subject <- defDataAdd(gen.subject,
                          varname = "Sex",
                          dist = "binary",
                          formula = 0.5)
gen.subject <- defDataAdd(gen.subject,
                          varname = "Age_T0",
                          dist = "normal",
                          formula = "6",
                          variance = 2^2)
gen.subject <- defDataAdd(gen.subject,
                          varname = "Age_T3",
                          dist = "nonrandom",
                          formula = "Age_T0 + 6",
                          variance = 2^2)
gen.subject <- defDataAdd(gen.subject,
```

```
                            varname = "Time_Outdoors_T0",
                            dist = "normal",
                            formula = "2",
                            variance = 2^2)
gen.subject <- defDataAdd(gen.subject,
                            varname = "Time_Outdoors_T3",
                            dist = "normal",
                            formula = "2",
                            variance = 1.5^2)


# ------------------------------------------------------
# 4. Generates subject-level data clustered within sites
# ------------------------------------------------------
# Generate the subject-level data
set.seed(282721)
dtSubject <- genCluster(dtSite,
                             cLevelVar = "Site_ID",
                             numIndsVar = "nSubjects",
                             level1ID = "Subject_ID")
dtSubject <- addColumns(gen.subject, dtSubject)

head(dtSubject)


# ---------------------------------------
# 5. Reshape data from wide to long format
# ---------------------------------------
# Reshape data from wide to long format
dt_long <- dtSubject %>%
  pivot_longer(
    cols = c(Age_T0, Age_T3, Time_Outdoors_T0, Time_Outdoors_T3),
    names_to = c(".value", "Visit"),
    names_pattern = "(.*)_(T.)"
  ) %>%
  mutate(Visit = ifelse(Visit == "T0", 1, 2)) %>%
  as.data.table()
```

```r
head(dt_long)


# ------------------------------------------------------
# 6. Define the Vitamin D level and deficiency variable
# ------------------------------------------------------
# Define the longitudinal Vitamin D level variable
gen_long <- defDataAdd(
  varname = "Vitamin_D_level",
  formula = "45 + 2*Sex - 0.5*Age + 1.5*Time_Outdoors + u_site + u_subject",
  variance = 14^2
)

# Add the longitudinal Vitamin D level variable
dtLong <- addColumns(gen_long, dt_long)

# Create the deficiency indicator variable
dtLong <- dtLong %>%
  # Indicator for deficiency
  mutate(Deficient = ifelse(Vitamin_D_level <= 50, 1, 0))
head(dtLong)


# ------------------------------------------------------
# 7. Save the simulated data
# ------------------------------------------------------
  save(dtLong, file = "dtLong.RData")
```

A.2 Appendix B. Synthetic Data Generation Code

This appendix provides the R code used to generate the synthetic datasets using the `synthpop` package, as described in Section 2.6.2.5 and summarized in Section 3.3.

The synthesis procedure used stratified synthesis by `Site_ID`, nested generation of `Subject_ID`, and variable-specific synthesis methods (CART for continuous variables, logistic regression for binary variables). Proper synthesis was applied throughout to incorporate uncertainty.

```r
# ----------------------------
# 1. Load necessary packages
```

```r
# ------------------------------
library(tidyverse)    # for data manipulation and visualization
library(synthpop)     # for generating synthetic data


# --------------------------------
# 2. Set seed for reproducibility
# --------------------------------
set.seed(282721)


# ----------------------------------------
# 3. Load the previously simulated dataset
# ----------------------------------------
load("dtLong.RData")      # Load original simulated data
dtLong <- as.data.frame(dtLong)

# Convert relevant columns to factors
dtLong <- dtLong %>%
  mutate(
    Sex = as.factor(Sex),
    Visit = as.factor(Visit),
    Deficient = as.factor(Deficient)
  )


# Check the first few rows of the dataset
head(dtLong, 10)


# ---------------------------------
# 4. Nested synthesis of Subject_ID
# ---------------------------------
# We synthesize Subject_ID separately to preserve clustering structure
# within sites. This ensure that dependencies between subjects
# within the same site are maintained.
synthetic_Subject_ID <- syn.nested(
  data = dtLong,
  y = dtLong$Subject_ID,  # Variable to synthesize
```

```r
  x = dtLong[, c("Site_ID", "u_site",   "nSubjects", "u_subject")], # Predictors
  # Predictors for nested synthesis
  xp = dtLong[, c("Site_ID", "u_site", "nSubjects", "u_subject")],
  method = "cart",
  proper = TRUE,
)

# Inspect the structure of the synthesized Subject_ID
str(synthetic_Subject_ID)

# Replace original Subject_ID with synthetic version
new_dtLong <- dtLong
new_dtLong$Subject_ID <- synthetic_Subject_ID$res


# ----------------------------------
# 5. Stratified synthesis by Site_ID
# ----------------------------------
# Stratifying ensures that site-level structure is preserved. The
# 'method' vector specifies the synthesis method for each column.
# Empty strings ("") indicate variables that are not synthesized
# during stratified synthesis. Specifically, Site_ID, u_site, nSubjects,
# and Subject_ID are left unchanged here.
# This for two complementary reasons:
# 1. Subject_ID has already been synthesized separately using
# `syn.nested`, preserving hierarchical structure.
# 2. Following the synthesization rule on **Predictor Matrix Adjustments**,
# identifiers such as Site_ID and Subject_ID are excluded as predictors
# for other variables. This avoids overfitting and redundancy,
# while correlations within sites and subjects are still preserved.
synthetic_data <- syn.strata(
  data = new_dtLong,
  strata = new_dtLong$Site_ID,  # Stratify by Site_ID
  method = c(
    "", "", "", "",     # Site_ID, u_site, nSubjects, Subject_ID (not synthesized)
    "cart", "logreg",   # u_subject (CART), Sex (logistic regression)
    "", "cart",         # Age -> CART
    "cart", "cart",     # Time_Outdoors, Vitamin_D_level -> CART
    "logreg"            # Deficient -> logistic regression
```

```
  ),
  minnumlevels = 2,        # Minimum number of levels for categorical variables
  proper = TRUE,           # Enable proper synthesis for valid inference
  drop.not.used = FALSE,   # Retain predictors not used for synthesis
  drop.pred.only = FALSE, # Keep predictors only used in synthesis
  seed = 282721            # Set seed for reproducibility
)$syn                      # Extract the synthesized dataset


# ----------------------------------
# 6. Inspect and save synthetic data
# ----------------------------------
View(synthetic_data)
save(synthetic_data, file = "synthetic_data.RData") # Save as RData for later use
```

A.3 Appendix C: Utility Assessment Code

This appendix provides the R code used to evaluate the **utility of the synthetic datasets**. As described in Section 2.6.2.6 and summarized in Section 3.4, both overall and variable-specific utility measures were computed to assess similarity between the synthetic and original datasets.

The pMSE and S_pMSE were used to evaluate overall and variable-specific utility. Histograms and density plots were generated to visualize differences between synthetic and original distributions.

```
# ----------------------------
# 1. Load necessary packages
# ----------------------------
library(tidyverse)     # for data manipulation and visualization
library(synthpop)      # for generating synthetic data


# --------------------------------
# 2. Load the datasets
# --------------------------------
load("dtLong.RData") # Loading the original simulated dataset

load("synthetic_data.RData") # Loading the synthetic dataset

# View the first few rows of the original dataset
```

```r
head(dtLong)

# View the first few rows of the synthetic dataset
head(synthetic_data)

# ----------------------------------------
# 3. Align data types
# ----------------------------------------
# To ensure consistency of datatypes between the synthesized
# and original data, we convert the relevant variables
# in both datasets to factors.
# Convert the relevant variables in dtLong to factors
# to match the synthesized data
dtLong <- dtLong %>%
  mutate(
    Site_ID = as.integer(Site_ID),
    Sex = as.factor(Sex),              # Convert to factor
    Visit = as.factor(Visit),          # Convert to factor
    Deficient = as.factor(Deficient)  # Convert to factor
  )

# Ensure data types align with synthesized data
synthetic_data <- synthetic_data %>%
  mutate(
    Site_ID = as.integer(Site_ID),
    Sex = as.factor(Sex),
    Visit = as.factor(Visit),
    Deficient = as.factor(Deficient)
  )


# Ensure the datasets are data frames, not data.table
synthetic_data <- as.data.frame(synthetic_data)
dtLong <- as.data.frame(dtLong)


# ----------------------------------
# 4. Verify structure alignment
# ----------------------------------
```

```r
# Checking the structure of both datasets to ensure that they match.
# Check if the structure matches after converting to data.frame
str(synthetic_data)
str(dtLong)

# Save dtLong as an RData file
save(dtLong, file = "dtLong.RData")
save(synthetic_data, file = "synthetic_data.RData")

# ------------------------------------------------------
# 5. Utility Comparison of Original and Synthesized Dataset
# ------------------------------------------------------
# We use the comparison and utility.gen function from the synthpop package
# to assess how similar the synthesized data is to the original data.
# Note: According to Nowok et al.(2016), a pSME or utility metric
# of less than or equal to 10 is generally considered an indicator of high utility.
# This means that the synthetic data closely approximates the original data
# in terms of preserving the relationships between variables.

# Compare variable-specific distribution between synthetic and original data
# using pMSE and S_pMSE (produces tables/plots for each variable).
compare(synthetic_data, dtLong)

# Calculates overall utility score (pMSE) between synthetic and original data
# using CART-based propensity score, without resampling.
u_ob_synth <- utility.gen(
  synthetic_data, dtLong,
  method = "cart",
  resamp.method = "none",  # no permutation/pairs resampling (faster, but no S_pMSE)
  print.flag = FALSE
)

u_ob_synth
```

A.4 Appendix D: Gold Standard Pooled Analysis Simulation Code

This appendix provides the complete R code used to perform the gold standard pooled analysis described in Section 3.5.1. The code reproduces the simulation of parameter estimates from the pooled original dataset, including model fitting and the calculation of bias, variance, and MSE across 1,000 repetitions.

All site-level data were analyzed jointly, and the script records estimates for both fixed and random effects at each iteration.

The full code is presented below to support transparency and reproducibility.

```
# ------------------------------
# 1. Load necessary packages
# ------------------------------
library(simstudy)  # Simulate realistic data for statistical modeling
library(lme4)      # Fit linear and generalized linear mixed-effects models
library(tidyverse) # Collection of packages for data manipulation and visualization
library(data.table)# Efficient data manipulation for large datasets
library(purrr)     # Functional programming tools for applying functions iteratively
library(furrr)     # Parallelized version of purrr functions for faster computation
library(knitr)     # Generate dynamic reports by embedding R code into documents


# ------------------------------------------------------------
# 2. Generating and Running the Simulation Study 1000 times
# ------------------------------------------------------------
# Set up parallel processing to run simulations across multiple sessions (cores)
plan(multisession)

# Modify the simulation function to extract additional components
run_simulation <- function(sim_id) {

  # Generate site data
  gen.site <- defData(varname = "u_site",
                      dist = "normal",
                      formula = 0,
                      variance = 3,
                      id = "Site_ID")
  gen.site <- defData(gen.site,
                      varname = "nSubjects",
                      formula = 80000,
                      variance = .3,
                      dist = "clusterSize")

  # Ensures different results by adjusting the seed per iteration
  set.seed(282721 + sim_id)
```

```
dtSite <- genData(8, gen.site, id = "Site_ID")


# Define subject-level data
gen.subject <- defDataAdd(varname = "u_subject",
                          dist = "normal",
                          formula = 0,
                          variance = 2)
gen.subject <- defDataAdd(gen.subject,
                          varname = "Sex",
                          dist = "binary",
                          formula = 0.5)
gen.subject <- defDataAdd(gen.subject,
                          varname = "Age_T0",
                          dist = "normal",
                          formula = "6",
                          variance = 2^2)
gen.subject <- defDataAdd(gen.subject,
                          varname = "Age_T3",
                          dist = "nonrandom",
                          formula = "Age_T0 + 6",
                          variance = 2^2)
gen.subject <- defDataAdd(gen.subject,
                          varname = "Time_Outdoors_T0",
                          dist = "normal",
                          formula = "2",
                          variance = 2^2)
gen.subject <- defDataAdd(gen.subject,
                          varname = "Time_Outdoors_T3",
                          dist = "normal",
                          formula = "2",
                          variance = 1.5^2)

# 3. Generate the subject-level data
set.seed(282721)
dtSubject <- genCluster(dtSite,
                        cLevelVar = "Site_ID",
                        numIndsVar = "nSubjects",
```

```
                          level1ID = "Subject_ID")
  dtSubject <- addColumns(gen.subject, dtSubject)

# 3. Reshape to long format and calculate Vitamin D level (same as before)
  dt_long <- dtSubject %>%
  pivot_longer(
    cols = c(Age_T0, Age_T3, Time_Outdoors_T0, Time_Outdoors_T3),
    names_to = c(".value", "Visit"), names_pattern = "(.*)_(T.)"
  ) %>%
  mutate(Visit = ifelse(Visit == "T0", 1, 2)) %>%
  as.data.table()

  # 5. Define the longitudinal variables and Vitamin D level
  gen_long <- defDataAdd(
    varname = "Vitamin_D_level",
    formula = "45 + 2*Sex - 0.5*Age + 1.5*Time_Outdoors + u_site + u_subject",
    variance = 14^2
  )



  # 6. Add the longitudinal variables to the data
  dtLong <- addColumns(gen_long, dt_long)

  # Define the deficiency variable
  dtLong <- dtLong %>%
    mutate(Deficient = ifelse(Vitamin_D_level <= 50, 1, 0))

  # Save dtLong as an RData File
  save(dtLong, file = "dtLong.RData")


  mod <- lmer(
    Vitamin_D_level ~ Sex + Age + Time_Outdoors + (1 | Site_ID/Subject_ID),
    REML = TRUE,
    data = dtLong
  )
```

```r
  # Extract fixed effects
  fixed_effects <- fixef(mod)

  # Extract random effects variances and residual variance
  random_effects <- as.data.frame(VarCorr(mod))
  residual_variance <- attr(VarCorr(mod), "sc")^2

  # Collect the results into a tibble
  tibble(
    sim_id = sim_id,
    `(Intercept)` = fixed_effects[1],
    Sex = fixed_effects["Sex"],
    Age = fixed_effects["Age"],
    Time_Outdoors = fixed_effects["Time_Outdoors"],
    `Subject_ID:Site_ID_variance` = random_effects[
      random_effects$grp == "Subject_ID:Site_ID", "vcov"
    ],
    `Site_ID_variance` = random_effects[
      random_effects$grp == "Site_ID", "vcov"
    ],
    Residual_variance = residual_variance
  )
}


# Run the simulation 1000 times
num_simulations <- 1000
results <- future_map_dfr(1:num_simulations, run_simulation, .id = "sim_id")

# Reset the plan to sequential processing when done
plan(sequential)

head(results, 10)

# --------------------
# 3. Result Summary
# --------------------
# Summarize results
```

```r
# Summarize results
summary_results <- results %>%
  summarise(
    mean_intercept = mean(`(Intercept)`),
    sd_intercept = sd(`(Intercept)`),
    var_intercept = var(`(Intercept)`),
    mean_sex = mean(Sex),
    sd_sex = sd(Sex),
    var_sex = var(Sex),
    mean_age = mean(Age),
    sd_age = sd(Age),
    var_age = var(Age),
    mean_time_outdoors = mean(Time_Outdoors),
    sd_time_outdoors = sd(Time_Outdoors),
    var_time_outdoors = var(Time_Outdoors),
    `mean_Subject_ID:Site_ID_variance` = mean(`Subject_ID:Site_ID_variance`),
    `sd_Subject_ID:Site_ID_variance` = sd(`Subject_ID:Site_ID_variance`),
    `var_Subject_ID:Site_ID_variance` = var(`Subject_ID:Site_ID_variance`),
    mean_Site_ID_variance = mean(`Site_ID_variance`),
    sd_Site_ID_variance = sd(`Site_ID_variance`),
    var_Site_ID_variance = var(`Site_ID_variance`),
    mean_Residual_variance = mean(Residual_variance),
    sd_Residual_variance = sd(Residual_variance),
    var_Residual_variance = var(Residual_variance)
  )


# Reshape to long format manually
summary_table <- data.frame(
  Parameter = c(
    "Intercept",
    "Sex",
    "Age",
    "Time Outdoors",
    "Subject_ID:Site_ID Variance",
    "Site_ID Variance",
    "Residual Variance"
  ),
```

```
  Mean = c(
    summary_results$mean_intercept,
    summary_results$mean_sex,
    summary_results$mean_age,
    summary_results$mean_time_outdoors,
    summary_results$`mean_Subject_ID:Site_ID_variance`,
    summary_results$mean_Site_ID_variance,
    summary_results$mean_Residual_variance
  ),

  SD = c(
    summary_results$sd_intercept,
    summary_results$sd_sex,
    summary_results$sd_age,
    summary_results$sd_time_outdoors,
    summary_results$`sd_Subject_ID:Site_ID_variance`,
    summary_results$sd_Site_ID_variance,
    summary_results$sd_Residual_variance
  ),

  Variance = c(
    summary_results$var_intercept,
    summary_results$var_sex,
    summary_results$var_age,
    summary_results$var_time_outdoors,
    summary_results$`var_Subject_ID:Site_ID_variance`,
    summary_results$var_Site_ID_variance,
    summary_results$var_Residual_variance
  )
)

# Create a table using kable
kable(
  summary_table,
  format = "markdown",
  col.names = c("Parameter", "Mean", "Standard Deviation", "Variance")
)
```

```r
##################################################
# 4. Performance Metrics Calculation
# ------------------------------------------------
# This section computes Bias, Variance, and MSE
# for each parameter based on the true parameter
# values and simulation results.
##################################################


# --- Define true parameter values ---
true_values <- c(
  Intercept = 45,
  Sex = 2,
  Age = -0.5,
  Time_Outdoors = 1.5,
  `Subject_ID:Site_ID_variance` = 2,
  Site_ID_variance = 3,
  Residual_variance = 14^2
)


# --- Construct summary table for Bias, Variance, and MSE ---
summary_metrics <- data.frame(

  # True parameter values
  True_Value = c(
    true_values["Intercept"],
    true_values["Sex"],
    true_values["Age"],
    true_values["Time_Outdoors"],
    true_values["Subject_ID:Site_ID_variance"],
    true_values["Site_ID_variance"],
    true_values["Residual_variance"]
  ),

  # Mean estimates across simulations
  Mean_Estimate = c(
    summary_results$mean_intercept,
    summary_results$mean_sex,
```

```
  summary_results$mean_age,
  summary_results$mean_time_outdoors,
  summary_results$`mean_Subject_ID:Site_ID_variance`,
  summary_results$mean_Site_ID_variance,
  summary_results$mean_Residual_variance
),

# Bias Calculation
Bias = c(
  summary_results$mean_intercept - true_values["Intercept"],
  summary_results$mean_sex - true_values["Sex"],
  summary_results$mean_age - true_values["Age"],
  summary_results$mean_time_outdoors - true_values["Time_Outdoors"],
  summary_results$`mean_Subject_ID:Site_ID_variance` -
    true_values["Subject_ID:Site_ID_variance"],
  summary_results$mean_Site_ID_variance - true_values["Site_ID_variance"],
  summary_results$mean_Residual_variance - true_values["Residual_variance"]
),

# Variance Calculation
Variance = c(
  var(results$`(Intercept)`),
  var(results$Sex),
  var(results$Age),
  var(results$Time_Outdoors),
  var(results$`Subject_ID:Site_ID_variance`),
  var(results$`Site_ID_variance`),
  var(results$Residual_variance)
),

# MSE Calculation
MSE = c(
  (summary_results$mean_intercept - true_values["Intercept"])^2 +
    var(results$`(Intercept)`),
  (summary_results$mean_sex - true_values["Sex"])^2 +
    var(results$Sex),
  (summary_results$mean_age - true_values["Age"])^2 +
    var(results$Age),
  (summary_results$mean_time_outdoors - true_values["Time_Outdoors"])^2 +
```

```
      var(results$Time_Outdoors),
    (summary_results$`mean_Subject_ID:Site_ID_variance` -
       true_values["Subject_ID:Site_ID_variance"])^2 +
      var(results$`Subject_ID:Site_ID_variance`),
    (summary_results$mean_Site_ID_variance - true_values["Site_ID_variance"])^2 +
      var(results$`Site_ID_variance`),
    (summary_results$mean_Residual_variance - true_values["Residual_variance"])^2 +
      var(results$Residual_variance)
  )
)

# View the final table
kable(
  summary_metrics,
  caption = "Summary of Bias, Variance, and MSE of Model Parameters"
)


# ------------------------------------------------------------
# 5. Save the summary_metrics
# ------------------------------------------------------------
write.csv(
  summary_metrics,
  "simulation_metrics_run_simulation_02.csv",
  row.names = FALSE
)
```

A.5 Appendix E: FSDA Simulation Code

This appendix contains the R code used for the FSDA simulations presented in Section 3.5.2. The code implements the site-level synthesis and model-fitting procedures, following the federated framework described in Section 2.3.

For each of the 1,000 repetitions, synthetic datasets were generated within each site, combined, and analyzed using the same model structure as in the gold standard pooled analysis.

The script records parameter estimates and computes summary performance metrics (bias, variance, and MSE) across all repetitions.

Finally, the appendix also includes the comparative evaluation of FSDA and gold standard pooled analysis results against the true parameter values from the data-generating mechanism.

The full R code is provided below for reproducibility and reference.

```
################################################################
# 1. Load Required Packages
# ---------------------------------------------------------
# These packages support data simulation, manipulation,
# modeling, and parallel computation
################################################################

library(simstudy)     # For simulating realistic datasets
library(lme4)         # For fitting linear mixed-effects models
library(tidyverse)    # Core packages for data manipulation and visualization
library(data.table)   # Fast and memory-efficient data handling
library(purrr)        # Functional programming (map functions)
library(furrr)        # Parallelized version of purrr for faster simulation
library(knitr)        # For producing formatted tables in reports
library(synthpop)     # For generating synthetic data
library(parallel)     # For accessing multiple CPU cores
library(kableExtra)   # For enhanced table display in reports



# ================================================
# 2. Set Seed and Load Simulated Data
# ================================================
# A fixed seed ensures full reproducibilty across runs.

set.seed(282721)

# Load the previously simulated dtLong data
load("dtLong.RData")
dtLong <- as.data.frame(dtLong)

# Convert categorical variables to factors
dtLong <- dtLong %>%
  mutate(
    Sex = as.factor(Sex),
    Visit = as.factor(Visit),
    Deficient = as.factor(Deficient)
```

```
  )


# Preview dataset
head(dtLong, 10)

# ==========================================
# 3. Parallel Processing Setup
# ==========================================
# Utilize all but one available CPU core to speed
# up simulation repetitions.

plan(multisession, workers = availableCores() - 1)



############################################################
# 4. Define True Parameter Values
# ----------------------------------------------------------
# These represent the known parameter values from
# the data generation mechanism against which
# simulation estimates will be evaluated.
############################################################

true_values <- c(
  Intercept = 45.0,
  Sex = 2.0,
  Age = -0.5,
  "Time_Outdoors" = 1.5,
  "Subject_ID:Site_ID_Variance" = 2,
  "Site_ID_Variance" = 3,
  "Residual_Variance" = 14^2
)



#####################################################################
# 5. Synthetic Data Simulation Function
# -------------------------------------------------------------------
```

```r
# This function performs one full iteration of the
# simulation process:
# 1. Synthesizes data using CART and logistic regression models.
# 2. Fits a mixed-effects model to the synthetic datasets.
# 3. Extracts key parameter estimates (fixed and random effects).
#########################################################################
simulate_once <- function(dtLong, sim_id) {

  # Step 1: Synthesize subject-level IDs using nested synthesis
  synthetic_Subject_ID <- syn.nested(
    data = dtLong,
    y = dtLong$Subject_ID,
    x = dtLong[, c("Site_ID", "u_site", "nSubjects", "u_subject")],
    xp = dtLong[, c("Site_ID", "u_site", "nSubjects", "u_subject")],
    method = "cart",
    proper = TRUE,
  )

  # Replace Subject_ID with synthesized version
  new_dtLong <- dtLong
  new_dtLong$Subject_ID <- synthetic_Subject_ID$res


  # Step 2: Stratified synthesis within each Site_ID
  synthetic_data <- syn.strata(
    data = new_dtLong,
    strata = new_dtLong$Site_ID,  # Stratify by Site_ID
    method = c(
      "", "", "", "",        # Site_ID, u_site, nSubjects, Subject_ID (not synthsized)
      "cart", "logreg",      # u_subject (CART), Sex (logistic regression)
      "", "cart",            # Age (CART)
      "cart", "cart",        # Time_Outdoors and Vitamin_D_level (CART)
      "logreg"               # Deficient (Logistic regression)
    ),
    minnumlevels = 2,        # Minimum number of levels for categorical variables
    proper = TRUE,           # Enable proper synthesis for valid inference
    drop.not.used = FALSE,   # Retain predictors not used for synthesis
    drop.pred.only = FALSE,  # Keep predictors only used in synthesis
    seed = 282721 + sim_id   # Set seed for reproducibility
```

```
  )$syn                          # Extract the synthesized data


  # Step 3: Fit linear mixed-effects model on the synthetic dataset
  mod <- lmer(
    Vitamin_D_level ~ Sex + Age + Time_Outdoors + (1 | Site_ID/Subject_ID),
    data = synthetic_data,
    REML = TRUE,
    control = lmerControl(
      optCtrl = list(maxfun = 1e6),      # Allow more iterations
      optimizer = "bobyqa",              # Use the 'bobyqa' optimizer
      check.conv.singular = "ignore"     # Relax singularity checks
    )
  )

  # Step 4: Extract fixed and random effect estimates
  fixed_effects <- fixef(mod)
  random_effects <- as.data.frame(VarCorr(mod))
  residual_variance <- attr(VarCorr(mod), "sc")^2

  # Step 5: Return results as a tidy tibble
  tibble(
    sim_id = sim_id,
    `(Intercept)` = fixed_effects[1],
    Sex = fixed_effects["Sex"],
    Age = fixed_effects["Age"],
    Time_Outdoors = fixed_effects["Time_Outdoors"],
    `Subject_ID:Site_ID_variance` = random_effects[
      random_effects$grp == "Subject_ID:Site_ID", "vcov"
    ],
    `Site_ID_variance` = random_effects[
      random_effects$grp == "Site_ID", "vcov"
    ],
    Residual_variance = residual_variance
  )

}
```

```r
################################################################
# 6. Batch processing and Logging
# -------------------------------------------------------------
# The following section handles automated batch
# execution of simulations:
# - Runs simulations in manageable batches ( 10 at a time).
# - Logs progress to a text file.
# - Saves intermediate batch results to CSV files.
# This ensures recovery in case of interruptions.
################################################################

# --- Define Parameters ---
batch_size <- 10
num_simulations <- 1000
results_dir <- "simulation_results_05"
log_file <- file.path(results_dir, "batch_processing_log.txt")
dir.create(results_dir, showWarnings = FALSE)

# --- Check for Existing Log and Resume from Last Completed Batch
if (file.exists(log_file)) {
  last_completed_line <- suppressWarnings(readLines(log_file))
  if (length(last_completed_line) == 0) {
    last_completed_batch <- 0
  } else {
    last_line <- tail(last_completed_line, 1)
    match <- regmatches(last_line, regexpr("\\d+ to \\d+", last_line))
    if (length(match) > 0) {
      last_completed_batch <- as.integer(strsplit(match, " to ")[[1]][2])
    } else {
      last_completed_batch <- 0
    }
  }
} else {
  last_completed_batch <- 0
}

# --- Function to Process Each Batch ---
process_batch <- function(start_id, end_id) {
  result <- future_map_dfr(start_id:end_id, ~ {
```

```
    set.seed(282721 + .x)
    tryCatch({
      simulate_once(dtLong, sim_id = .x)
    }, error = function(e) {
      message("Error in simulation ", .x, ": ", e$message)
      return(NULL)
    })
  }, .options = furrr_options(seed = TRUE))


  # Save results and log completion
  if (!is.null(result)) {
    csv_file_path <- file.path(
      results_dir,
      paste0("simulation_batch_", start_id, "_to_", end_id, ".csv")
    )
    write.csv(result, file = csv_file_path, row.names = FALSE)

    cat(
      paste("Batch from", start_id, "to", end_id, "completed\n"),
      file = log_file,
      append = TRUE
    )
  }
}


# --- Main Batch Processing Loop ---
start_simulation <- last_completed_batch + 1
end_simulation <- min(start_simulation + batch_size - 1, num_simulations)

while (start_simulation <= num_simulations) {
  tryCatch(
    {
      process_batch(start_simulation, end_simulation)
    },
    error = function(e) {
      message(
        "Error processing batch ",
```

```
        start_simulation, " to ", end_simulation, ": ", e$message
      )
      cat(
        paste(
          "Error in batch from",
          start_simulation, "to", end_simulation, ": ", e$message, "\n"
        ),
        file = log_file,
        append = TRUE
      )
    }
  )

  # Update batch range
  start_simulation <- end_simulation + 1
  end_simulation <- min(
    start_simulation + batch_size - 1,
    num_simulations
  )
}



############################################################
# 7. Combine and Summarize Results
# ----------------------------------------------------------
# This section combines all saved simulation result files
# and computes summary statistics (mean, SD, variance) for
# each model parameter.
############################################################

# --- Ensure Sequential Execution (for Controlled Reproducibility) ---
plan(sequential)

# --- Define Directory Containing Simulation Batch Files ---
# Each batch file contains parameter estimates from multiple
# repetitions of the simulation study.
all_results <- list.files(
  path = results_dir,
```

```
  pattern = "simulation_batch_.*.csv",
  full.names = TRUE
) %>%
  map_dfr(~ read.csv(.x, check.names = FALSE))

# --- Preview Combined Results (first 10) ---
head(all_results, 10)



# ============================================
# Summarize Results Across All Simulations
# ============================================

summary_results <- all_results %>%
  summarise(
    # --- Fixed Effects ---
    mean_intercept = mean(`(Intercept)`),
    sd_intercept = sd(`(Intercept)`),
    var_intercept = var(`(Intercept)`),

    mean_sex = mean(Sex),
    sd_sex = sd(Sex),
    var_sex = var(Sex),

    mean_age = mean(Age),
    sd_age = sd(Age),
    var_age = var(Age),

    mean_time_outdoors = mean(Time_Outdoors),
    sd_time_outdoors = sd(Time_Outdoors),
    var_time_outdoors = var(Time_Outdoors),

    # --- Random Effects ---
    `mean_Subject_ID:Site_ID_variance` = mean(`Subject_ID:Site_ID_variance`),
    `sd_Subject_ID:Site_ID_variance` = sd(`Subject_ID:Site_ID_variance`),
    `var_Subject_ID:Site_ID_variance` = var(`Subject_ID:Site_ID_variance`),

    mean_Site_ID_variance = mean(`Site_ID_variance`),
    sd_Site_ID_variance = sd(`Site_ID_variance`),
```

```
    var_Site_ID_variance = var(`Site_ID_variance`),

    # --- Residual Variance ---
    mean_Residual_variance = mean(Residual_variance),
    sd_Residual_variance = sd(Residual_variance),
    var_Residual_variance = var(Residual_variance)
  )


##########################################################
# 8. Results Table
# -------------------------------------------------------
# This section presents the summarized results in
# tabular form, showing the mean, standard deviation,
# and variance for each parameter.
##########################################################

# --- Construct Summary Table ---
summary_table <- data.frame(
  Parameter = c(
    "Intercept",
    "Sex",
    "Age",
    "Time_Outdoors",
    "Subject_ID:Site_ID_Variance",
    "Site_ID_Variance",
    "Residual_Variance"
  ),

  Mean = c(
    summary_results$mean_intercept,
    summary_results$mean_sex, summary_results$mean_age,
    summary_results$mean_time_outdoors,
    summary_results$`mean_Subject_ID:Site_ID_variance`,
    summary_results$mean_Site_ID_variance,
    summary_results$mean_Residual_variance
  ),

  SD = c(
```

```
    summary_results$sd_intercept,
    summary_results$sd_sex,
    summary_results$sd_age,
    summary_results$sd_time_outdoors,
    summary_results$`sd_Subject_ID:Site_ID_variance`,
    summary_results$sd_Site_ID_variance,
    summary_results$sd_Residual_variance
  ),

  Variance = c(
    summary_results$var_intercept,
    summary_results$var_sex,
    summary_results$var_age,
    summary_results$var_time_outdoors,
    summary_results$`var_Subject_ID:Site_ID_variance`,
    summary_results$var_Site_ID_variance,
    summary_results$var_Residual_variance)
)

# ================================================
# Displays the Results Table
# ================================================

kable(
  summary_table,
  format = "markdown",
  col.names = c("Parameter", "Mean", "Standard Deviation", "Variance"),
  caption = "Summary of Simulation Results Across All Parameters"
)


##########################################################
# 9. Bias, Variance, and MSE
# --------------------------------------------------
# This section calculates Bias, Variance, and MSE for
# each parameter based on the true parameter values
# and the results obtained across all simulation runs.
##########################################################
```

```r
# --- Define True Parameter Values
true_values <- c(
  Intercept = 45,
  Sex = 2,
  Age = -0.5,
  Time_Outdoors = 1.5,
  `Subject_ID:Site_ID_variance` = 2,
  Site_ID_variance = 3,
  Residual_variance = 14^2
)


# ========================================================
# Calculates Bias, Variance, and MSE
# ========================================================

summary_metrics <- data.frame(

  # --- True Values
  True_Value = c(
    true_values["Intercept"],
    true_values["Sex"],
    true_values["Age"],
    true_values["Time_Outdoors"],
    true_values["Subject_ID:Site_ID_variance"],
    true_values["Site_ID_variance"],
    true_values["Residual_variance"]
  ),

  # --- Mean Estimates (Averaged Across Simulation) ---
  Mean_Estimate = c(
    summary_results$mean_intercept,
    summary_results$mean_sex,
    summary_results$mean_age,
    summary_results$mean_time_outdoors,
    summary_results$`mean_Subject_ID:Site_ID_variance`,
    summary_results$mean_Site_ID_variance,
    summary_results$mean_Residual_variance
```

```
),


# --- Bias Calculation ---
Bias = c(
  summary_results$mean_intercept - true_values["Intercept"],
  summary_results$mean_sex - true_values["Sex"],
  summary_results$mean_age - true_values["Age"],
  summary_results$mean_time_outdoors - true_values["Time_Outdoors"],
  summary_results$`mean_Subject_ID:Site_ID_variance` -
    true_values["Subject_ID:Site_ID_variance"],
  summary_results$mean_Site_ID_variance - true_values["Site_ID_variance"],
  summary_results$mean_Residual_variance - true_values["Residual_variance"]
),

# --- Variance Calculation (Across Simulation Replicates) ---
Variance = c(
  var(all_results$`(Intercept)`),
  var(all_results$Sex),
  var(all_results$Age),
  var(all_results$Time_Outdoors),
  var(all_results$`Subject_ID:Site_ID_variance`),
  var(all_results$`Site_ID_variance`),
  var(all_results$Residual_variance)
),

# --- MSE Calculation ---
MSE = c(
  (summary_results$mean_intercept - true_values["Intercept"])^2 +
    var(all_results$`(Intercept)`),
  (summary_results$mean_sex - true_values["Sex"])^2 +
    var(all_results$Sex),
  (summary_results$mean_age - true_values["Age"])^2 +
    var(all_results$Age),
  (summary_results$mean_time_outdoors - true_values["Time_Outdoors"])^2 +
    var(all_results$Time_Outdoors),
  (summary_results$`mean_Subject_ID:Site_ID_variance` -
     true_values["Subject_ID:Site_ID_variance"])^2 +
    var(all_results$`Subject_ID:Site_ID_variance`),
```

```
    (summary_results$mean_Site_ID_variance - true_values["Site_ID_variance"])^2 +
      var(all_results$`Site_ID_variance`),
    (summary_results$mean_Residual_variance - true_values["Residual_variance"])^2 +
      var(all_results$Residual_variance)
  )
)


# ============================================
# Displays the Summary Table
# ============================================
kable(
  summary_metrics,
  caption = "Summary of Bias, Variance, and MSE Across Parameters"
)



##########################################################################
# 10. Comparison of Pooled Analysis and FSDA Results with True Values
# ------------------------------------------------------------------------
# This section compares parameter estimates from the gold standard pooled
# analysis and FSDA with the known true values.
##########################################################################


# --- Read the Summary Metrics Files ---
# These files contain Bias, Variance, and MSE results from both analyses.
metrics_run_simulation <- read.csv(
  "simulation_metrics_run_simulation_02.csv"
  )
metrics_run_simulation_synthpop <- read.csv(
  "simulation_metrics_run_simulation_synthpop_02.csv"
  )


# --- Define the True Parameter Values ---
true_values <- c(
  Intercept = 45.0,
```

```
  Sex = 2.0,
  Age = -0.5,
  "Time_Outdoors" = 1.5,
  "Subject_ID:Site_ID_variance" = 2,
  "Site_ID_variance" = 3,
  "Residual_variance" = 14^2
)


# --- Define Desired Parameter Order for Presentation ---
desired_order <- c(
  "Intercept",
  "Sex",
  "Age",
  "Time_Outdoors",
  "Subject_ID:Site_ID_variance",
  "Site_ID_variance",
  "Residual_variance"
)

# ======================================================
# Create Comparison Data Frame
# ======================================================


# --- Pooled (Simulated) Data Results ---
comparison_run_simulation <- data.frame(
  Parameter = desired_order,
  True_Value = true_values[desired_order],
  "Mean_Estimate (Simulated Data)" = metrics_run_simulation$Mean_Estimate,
  "Bias (Simulated Data)" = metrics_run_simulation$Bias,
  "Variance (Simulated Data)" = metrics_run_simulation$Variance,
  "MSE (Simulated Data)" = metrics_run_simulation$MSE
)

# --- Federated Synthetic Data (FSDA) Results ---
comparison_run_simulation_synthpop <- data.frame(
  Parameter = desired_order,
  "Mean_Estimate (Synthetic Data)" = metrics_run_simulation_synthpop$Mean_Estimate,
```

```
  "Bias (Synthetic Data)" = metrics_run_simulation_synthpop$Bias,
  "Variance (Synthetic Data)" = metrics_run_simulation_synthpop$Variance,
  "MSE (Synthetic Data)" = metrics_run_simulation_synthpop$MSE
)

# ========================================================
# Merging the Two Data Frames for Side-by-Side Comparison
# ========================================================

comparison_df <- merge(
  comparison_run_simulation,
  comparison_run_simulation_synthpop,
  by = "Parameter"
)

# Reorders the comparison data frame based on the desired order
comparison_df$Parameter <- factor(comparison_df$Parameter, levels = desired_order)
comparison_df <- comparison_df[order(comparison_df$Parameter), ]

# Resets row numbers to reflect the correct order
rownames(comparison_df) <- NULL

# ================================================
# Display of the Comparison Table
# ================================================

# Selects and rename columns for clean presentation
comparison_df %>%
  select(
    Parameter,
    True_Value,
    starts_with("Mean_Estimate"),
    starts_with("Bias"),
    starts_with("Variance"),
    starts_with("MSE")
  ) %>%
  rename(
    "Mean_Estimate (Simulated Data)" = "Mean_Estimate..Simulated.Data.",
    "Bias (Simulated Data)" = "Bias..Simulated.Data.",
```

```
    "Variance (Simulated Data)" = "Variance..Simulated.Data.",
    "MSE (Simulated Data)" = "MSE..Simulated.Data.",
    "Mean_Estimate (Synthetic Data)" = "Mean_Estimate..Synthetic.Data.",
    "Bias (Synthetic Data)" = "Bias..Synthetic.Data.",
    "Variance (Synthetic Data)" = "Variance..Synthetic.Data.",
    "MSE (Synthetic Data)" = "MSE..Synthetic.Data."
  ) %>%
  kable(
    "html",
    caption = "Comparison of Simulated and Synthetic Data Results with the True Values"
  ) %>%
  kable_styling(
    full_width = TRUE,
    Position = "center"
  )
```

A.6 Appendix F: Disclosure Risk Assessment Code

This appendix presents the R implementation used to evaluate disclosure risks for the synthetic data generated in this thesis. The assessment followed the framework of Raab et al. (2024), applying both **identity and attribute disclosure metrics** using the `multi.disclosure()` function from the `synthpop` package.

```
# ------------------------------
# 1. Load necessary packages
# ------------------------------
library(dplyr)        # data manipulation and transformation
library(synthpop)     # generating and evaluating synthetic data
library(data.table)   # data manipulation and handling of datasets

# --------------------------------
# 2. Load the datasets
# --------------------------------
load("dtLong.RData")          # Loads the original simulated dataset

load("synthetic_data.RData")  # Loads the synthetic dataset


#########################################################
# 3. Align data types
```

```r
# ----------------------------------------------------
# To ensure consistency of datatypes between
# the synthesized and original data, we convert
# the relevant variables in both datasets to factors.
######################################################
# Converts the relevant variables in dtLong to factors
# to match the synthesized data
dtLong <- dtLong %>%
  mutate(
    Site_ID = as.integer(Site_ID),
    Sex = as.factor(Sex),              # Convert to factor
    Visit = as.factor(Visit),          # Convert to factor
    Deficient = as.factor(Deficient)  # Convert to factor
  )

# Ensures data types align with synthesized data
synthetic_data <- synthetic_data %>%
  mutate(
    Site_ID = as.integer(Site_ID),
    Sex = as.factor(Sex),
    Visit = as.factor(Visit),
    Deficient = as.factor(Deficient)
  )


# Converts data to data.table format for efficient filtering
synthetic_data_0 <- as.data.table(synthetic_data_0)
dtLong <- as.data.table(dtLong)


# Convert Site_ID to integer in both datasets
synthetic_data$Site_ID <- as.integer(synthetic_data$Site_ID)
dtLong$Site_ID <- as.integer(dtLong$Site_ID)

# Converts Subject_ID to integer in both datasets
# If Subject_IDs are numeric strings, as.integer will work
synthetic_data$Subject_ID <- as.integer(synthetic_data$Subject_ID)
dtLong$Subject_ID <- as.integer(dtLong$Subject_ID)
```

```r
# ---------------------------------
# 4. Verify structure alignment
# ---------------------------------
# Checks if the structure of both datasets matches after
# converting to data.table
str(synthetic_data)
str(dtLong)



# ------------------------------------------------------------
# 5. Site-wise Disclosure Risk Assessment
# ------------------------------------------------------------
# Extracts unique sites
site_ids <- unique(synthetic_data$Site_ID)

# Runs disclosure risk analysis per site
map(site_ids, function(site) {
  syn_risk <- multi.disclosure(
    object = synthetic_data[Site_ID == site],
    data   = dtLong[Site_ID == site],
    print.flag = FALSE,
    plot = FALSE,
    keys = c("Sex", "Visit", "Age", "Time_Outdoors"),
    targets = c("Subject_ID", "Vitamin_D_level", "Deficient"),
    ngroups_keys = c(0, 0, 5, 5),    # groups continuous keys into 5 bins
    ngroups_targets = c(0, 5, 0),    # groups continuous target into 5 bins
    ident.meas = "repU",             # identity disclosure measure
    attrib.meas = "DiSCO"            # attribute disclosure measure
  )

  # prints the site results
  cat("\nResults for Site_ID:", site, "\n")
  print(syn_risk, to.print = "ident")   # Identity disclosure (UiO, RepU)
  print(syn_risk, to.print = "attrib")  # Attribute disclosure (Dorig, DiSCO)
})
```

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